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Computer Science Research Report

When Is a Time-Series Change Local? MetaShift-Bench for Selective Scope Auditing

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When Is a Time-Series Change Local?

MetaShift-Bench for Selective Scope Auditing

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Abstract

When a monitored time series changes, a target-only detector can identify the break but cannot determine whether it is local to that target or shared with a comparison network. This distinction matters because a local anomaly and a regional common shock call for different follow-up actions, while many real monitoring records lack reliable scope labels. We formulate selective scope auditing as an information-constrained problem: at a stated error tolerance, which cases can an observation channel answer, and when should it abstain?

MetaShift-Bench holds the target path exactly fixed across paired local and shared synthetic worlds and varies only donor-side participation. Its primary target-fixed experiment contains 360 held-out components, 230,400 matched pairs, and a complete $5 \times 4 \times 2^5$ adverse-condition grid. In the finite predeclared policy set, target-only information answers no held-out cases below 50% tolerated error. Comparative information also answers none through $\alpha = .10$, then reaches 60.4% coverage at $\alpha = .20$ with 19.5% observed conditional error. A structural interval-separation certificate answered 89,997 of 230,400 matched pairs (39.1%), with no observed errors among 179,994 answered scope arms. It recovered 97.7% of the pairs deemed answerable by the predeclared simulation-information oracle and abstained on the remaining 140,403 pairs (60.9%), including all 46,080 observationally indistinguishable $q = 0$ negative controls. These finite-grid results hold only under the bounded synthetic generator and do not constitute a population-risk guarantee or a deployable real-network certificate.

We prove the target-only impossibility statement under matched, balanced target laws, derive the availability-normalized partial-scope residual gap, and give a sufficient condition for structural abstention. A separate AQS deployment audit measures comparison availability and abstention in real public records; it supplies neither scope labels nor mechanism truth. In its independent stable-regime benchmark, cross-site methods substantially exceed single-series baselines in local-versus-shared ranking, but MetaShift shows no confidence-supported aggregate advantage over standard synthetic control. The central result is therefore an evaluation framework for answering only information-supported scope questions.

Keywords: local-versus-shared change; time-series auditing; selective prediction; cross-site comparison; structural abstention.

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1 Introduction

A PM_{2.5} monitor can show an abrupt change near a reported Method Code transition. If nearby monitors move at the same time, the follow-up question is very different from the question raised by an isolated change. A target-only detector can identify a numerical break, but it cannot determine whether that break is local to the target or shared across the comparison network. Forcing a scope label when the necessary comparison is unavailable can misdirect a technical investigation just as readily as missing a change.

This is an information-constrained inference problem in computer science: selective prediction over time-series observations with explicitly limited channels. The central question is not “which estimator wins?” It is:

Given a specified observation channel and an allowed scope-error rate, what fraction of scope questions can a system answer, and when must it abstain?

We call this *information-constrained selective scope auditing*. Here, *scope* means whether a change is specific to the target or shared by its declared comparison network. It must be separated from detection and mechanism (Table 1).

Table 1: Three distinct questions raised by a changing monitored time series.

Question	What it asks	Minimum evidence needed
Detection	Did the target path change?	Target time series.
Scope	Is the change local or shared with the comparison network?	Target series plus a qualified cross-site comparison.
Mechanism	Why did the change occur?	Technical records, domain metadata, and human review in addition to a comparison.

A reported AQS Method Code transition is useful as an observable metadata anchor, but it is not evidence that a device was replaced, failed, or biased. That distinction makes the real-data audit a test of comparison availability and principled abstention, rather than a source of unobserved scope or mechanism labels.

MetaShift-Bench supplies a deliberately target-fixed synthetic test of the scope question. Every matched pair has exactly identical target observations in the local and shared arms; only donor participation changes. Consequently, a target-only policy cannot distinguish balanced local from shared labels in the stipulated experiment, while a policy receiving target and donor channels may do so. The benchmark measures the held-out coverage achieved by predeclared policies at stated answered-case error tolerances.

The primary design extends a binary local-versus-regional endpoint into a donor-participation continuum. It crosses nominal participation $q \in \{0, .25, .50, .75, 1\}$ with signal strength and adverse comparison conditions. A bounded structural certificate answers only when availability-normalized score intervals separate under declared generator envelopes. A nonpositive margin is retained as a required abstention region, not hidden inside a weak classifier score.

The report asks five linked research questions:

1. **RQ1:** How much scope-answerability coverage do target-only and comparative channels achieve at prespecified error tolerances?
2. **RQ2:** When does a structural certificate answer or abstain along the partial-scope boundary?
3. **RQ3:** How do cross-site and single-series methods compare, and does MetaShift have a confidence-supported advantage over standard synthetic control?
4. **RQ4:** How often can a qualified cross-site audit be formed in the AQS deployment inventory, and what dispositions result?

5. **RQ5:** What do interval calibration and contextual evidence reveal about the limits of real-data auditing?

Our contributions are fourfold:

- C1. Formulate** a target-fixed benchmark that measures selective local-versus-shared scope answerability by observation channel.
- C2. Derive** an availability-normalized partial-scope residual identity and a bounded interval-separation condition for structural abstention.
- C3. Evaluate** predeclared confidence and certificate policies on a complete adverse-condition grid, retaining failures and identity controls.
- C4. Audit** an independent AQS deployment inventory to measure comparison availability, diagnostic dispositions, and the limits of metadata-based interpretation.

The framework builds on established selective classification, information refinement, change-point detection, and synthetic control. Its contribution is their bounded joint evaluation for an explicitly target-fixed scope question.

2 Problem formulation

2.1 Three questions and observation channels

For one target time series, a target-only channel sees the target path itself; a comparative channel adds donor paths and their availability; an external channel would additionally contain admissible station or maintenance records. Let $Y \in \{L, S\}$ denote a stipulated synthetic scope label, where L is target-local and S is shared with comparative donors. The conceptual nested channels are

$$O_T = X, \quad O_C = (X, D), \quad O_E = (X, D, E).$$

Here X is the declared target path, D contains donor paths, fixed donor weights, and availability indicators, and E denotes admissible external records. The synthetic experiment supplies constructed labels only for detection and scope. It contains neither an empirical O_E channel nor mechanism labels.

A selective policy $\phi = (g, s)$ maps an allowed observation channel and auxiliary randomness to a scope prediction g and an answer indicator $s \in \{0, 1\}$. Its population answerability frontier at tolerated conditional scope error α is

$$\Gamma_{\mathcal{I}}(\alpha) = \sup_{\substack{\phi \text{ } \mathcal{I}\text{-measurable} \\ \Pr(s_\phi=1)>0 \\ \Pr(g_\phi(O) \neq Y | s_\phi=1) \leq \alpha}} \Pr(s_\phi = 1).$$

The supremum is defined as zero if no positive-coverage policy meets the constraint. This is an inverse selective risk-coverage objective specialized to a scope label; it is not a new generic risk-coverage definition [22, 23].

Table 2: Notation for the target-fixed partial-scope construction.

Symbol	Meaning
a_{jt} and $\tilde{w}_{jt}$	Donor j 's availability and its availability-normalized weight on date t .
λ_{jt} and q_t	Declared donor participation and realized weighted participation on date t .
h_t and H	Injected target schedule and its constant-schedule magnitude.
A	Post-minus-pre mean residual score used for the scope decision.
B_L, B_S, κ	Declared local/shared error envelopes and structural separation margin.

The tolerance α is the largest permitted error rate among cases that a policy answers. Abstained cases do not enter this conditional error. In the experiment, the theoretical frontier is not directly observed. Instead, a finite set of predeclared policies is evaluated on held-out components.

For nested channels, the *Scope Answerability Gain* is

$$\Delta_{\mathcal{I} \rightarrow \mathcal{J}}(\alpha) = \text{SAG}_{\mathcal{I} \rightarrow \mathcal{J}}(\alpha) = \Gamma_{\mathcal{J}}(\alpha) - \Gamma_{\mathcal{I}}(\alpha).$$

The benchmark-specific name does not make its nonnegativity a new decision-theoretic principle. If the richer channel can discard its additional coordinates, its policy set includes the poorer-channel policy set, a Blackwell-style information-refinement argument [14, 15].

2.2 Target-only impossibility and finite-policy estimation

Proposition 1 (target-fixed impossibility). Suppose the synthetic labels are balanced and the matched construction satisfies

$$\Pr(Y = L) = \Pr(Y = S) = \frac{1}{2}, \quad \mathcal{L}(X \mid Y = L) = \mathcal{L}(X \mid Y = S).$$

If selection and prediction use only X and auxiliary randomness that is conditionally independent of Y given X , then every positive-coverage target-only policy has conditional scope error $1/2$. Therefore

$$\Gamma_T(\alpha) = 0 \quad \text{for } \alpha < \frac{1}{2}.$$

Proof. Conditioning on a target-only answer does not change the balanced posterior label probability because the accepted target law is the same under both labels. Either binary prediction is consequently wrong with probability $1/2$. $\square$

This is not a generic impossibility theorem for time series. It is the correctness argument for the benchmark’s matched construction: it would fail if target-only input leaked a pair identifier, donor availability, participation value, label schedule, generator truth, or label-dependent randomness.

Observation 1 (information refinement). When O_C contains O_T under the same joint law, label space, loss, and policy class,

$$\Gamma_C(\alpha) \geq \Gamma_T(\alpha).$$

The result follows by implementing every target-only policy after discarding D . It neither implies strict improvement nor validates any finite-sample comparative classifier.

The experiment estimates neither unrestricted population frontier. It reports the held-out finite-policy envelope

$$\hat{\Gamma}_{\mathcal{I}}^{\mathcal{P}}(\alpha) = \max_{\substack{\phi \in \mathcal{P} \\ \hat{R}_{\phi} \leq \alpha}} \hat{C}_{\phi},$$

with zero when no positive-coverage frozen policy qualifies. Here $\mathcal{P}$ contains the target-only forced rule, comparative forced rule, four separately calibration-selected confidence rules, and the specified synthetic-design-assisted certificate. Thus every displayed result is a descriptive held-out envelope for the frozen generator and policy set, not a population optimum, conditional-risk guarantee, or post-evaluation policy choice.

2.3 Target-fixed partial-scope construction

The primary construction is performed on the analysis scale $z = f(y) = \log(1 + \max(y, 0))$. Let $w_j \geq 0$ be fixed donor weights and a_{jt} denote donor availability. On every retained date,

$$\tilde{w}_{jt} = \frac{w_j a_{jt}}{\sum_k w_k a_{kt}}, \quad q_t = \sum_j \tilde{w}_{jt} \lambda_{jt},$$

where $\lambda_{jt} \in [0, 1]$ is declared donor participation. For positive post-anchor schedule h_t , the paired worlds are

$$z_t^L = z_t + h_t, \quad z_{jt}^L = z_{jt}, \quad z_t^S = z_t + h_t, \quad z_{jt}^S = z_{jt} + \lambda_{jt} h_t.$$

The target path is exactly identical in both worlds. With fixed offset b ,

$$r_t = z_t - \sum_j \tilde{w}_{jt} z_{jt} - b, \quad A = \text{mean}_{W_{\text{post}}} r_t - \text{mean}_{W_{\text{pre}}} r_t.$$

Proposition 2 (construction-specific residual identity). With arm-invariant availability, retained dates, weights, preprocessing, and pre-anchor offset,

$$r_t^L - r_t = h_t, \quad r_t^S - r_t = (1 - q_t)h_t.$$

Intuitively, a local injection remains fully in the target residual, while a shared injection is partly canceled by donors. The residual gap is therefore realized donor participation times the injected signal. For a constant post schedule $h_t = H$, the score-center separation is

$$A^L - A^S = \overline{q_t H}.$$

For a variable schedule the correct expression is $\overline{q_t h_t}$, not $\bar{q} \bar{h}$. This is exact algebra for the stipulated analysis-scale construction, not a generic identifiability claim. A raw-scale additive field does not preserve this identity after the nonlinear transformation and is treated only through a conservative leakage envelope.

2.4 Structural Answerability Certificate

The certificate is a synthetic-design-information-assisted policy, not a deployable real-network rule. Denoted O_C^+ , it receives generated participation and bounded-error quantities unavailable to a deployed monitoring system. For its efficiency diagnostic alone, the predeclared simulation-information oracle uses the same interval rule after replacing the conservative lower gap $G_{\min}$ with the generator-known realized gap $G = \overline{q_t h_t}$. It is an accounting reference, not a theoretical optimum, population upper bound, or deployment channel. Suppose its local and shared scores obey

$$|A^L - H| \leq B_L, \quad |A^S - (H - G)| \leq B_S,$$

and all scored retained dates satisfy $q_t \geq q_{\min}$ and $h_t \geq H_{\min} > 0$. Let

$$G_{\min} = q_{\min} H_{\min}, \quad \kappa = G_{\min} - (B_L + B_S).$$

Proposition 3 (interval-separation certificate). If $\kappa > 0$, the local and shared score intervals are disjoint. Every threshold in

$$(H - G_{\min} + B_S, H - B_L)$$

separates them under the stated bounds. The implementation uses their midpoint,

$$\tau_{\text{cert}} = H - \frac{G_{\min}}{2} + \frac{B_S - B_L}{2},$$

answers only if $\kappa > 0$, and predicts local if $A > \tau_{\text{cert}}$.

Proof. The upper shared bound is $H - G_{\min} + B_S$ and the lower local bound is $H - B_L$; their strict ordering is exactly $\kappa > 0$. $\square$

The theorem requires bounded noise, fixed retained-date semantics, valid availability normalization, modeled donor mismatch, bounded contamination, and a clipping-aware raw-field bound. Nonfinite values, changed dates, unmodeled transformations, fewer than three donors, or an envelope violation invalidate the certificate. When $\kappa \leq 0$, it must abstain. A nominal midpoint $H - G_{\min}/2$ is not generally safe under asymmetric bounds; the asymmetric term above is necessary.

3 Related work

3.1 Information and selective prediction

Blackwell’s comparison of statistical experiments formalizes when one observation can reproduce another decision problem by discarding or garbling information [14, 15]. The nested-channel observation used here is a direct specialization: a comparative policy can always discard donor information and imitate a target-only policy. It does not imply that any fitted rule improves in a finite sample or that comparative records are valid in an operational network.

Selective classification and reject-option learning already study the risk–coverage trade-off, optimal coverage at a chosen risk, and the cost of abstention [10, 18, 22, 23, 27]. Hierarchical selective classification permits a less-specific answer when a fine label is unjustified [29]. Conformal and predictive-inference work clarifies which assumptions are needed before interval statements are calibrated [6, 7, 35, 39, 48]. MetaShift-Bench uses these established ideas for a different selected object: a local-versus-shared scope judgment under explicitly contrasting observation channels.

3.2 Local/shared changes, residual signatures, and weak labels

Multivariate change-point work separates common and idiosyncratic components or tests common breaks in panels [5, 8]. Single-series procedures such as CUSUM, PELT, and multiscale segmentation remain useful for detecting a break, but cannot alone establish whether it is target-local or shared [25, 32, 41]. Weak-supervision research likewise distinguishes proxy labels from known truth [17, 24, 36, 38]. The synthetic scope labels in this report are defined by construction; a reported AQS Method Code is only a metadata anchor.

Residual signatures, analytical redundancy, fault distinguishability, and sensor placement are established in model-based diagnosis [19, 33, 40]. Climate-record homogenization similarly uses pairwise references and station histories to identify inhomogeneities [37, 49]. These literatures motivate the comparison logic, but their physical models and documented mechanisms are not assumed by this benchmark.

3.3 Counterfactual monitoring and public metadata

Synthetic control supplies donor-weighted counterfactuals and in-space placebo logic [1–3]. Generalized, augmented, and synthetic difference-in-differences variants address other panel

Table 3: Research traditions that inform the benchmark and the role retained here.

Research tradition	Established object	Role in MetaShift-Bench
Selective prediction [18, 27]	Risk-coverage and reject-option policies.	Measures answered-case coverage for a scope question; does not rename the generic objective.
Common/idiosyncratic change analysis [5, 8]	Shared and unit-specific movement in multivariate series.	Holds the target fixed while varying donor participation in paired worlds.
Synthetic control and DiD [2, 4]	Donor-weighted comparisons under explicit assumptions.	Uses pre-anchor fitting and falsification diagnostics, not policy-style causal identification.
Residual diagnosis and homogenization [19, 37]	Signatures, reference series, and documented records.	Uses a bounded synthetic separation condition and keeps mechanism interpretation external.
Monitoring metadata [44, 47]	Administrative fields and technical documentation.	Uses Method Code as a reproducible audit anchor, not physical ground truth.

structures and regularization choices [4, 11, 50]; modern difference-in-differences also emphasizes assumptions and treatment-timing heterogeneity [16, 28]. Moving-block bootstrap methods address serial dependence in time-series resampling [34], while multiple-testing and post-selection literature warns against overinterpreting screened findings [12, 13].

EPA publishes AQS file formats, Method Code and POC references, and monitoring comparability resources [43–47]. Collocated-monitor studies show why configuration-specific comparison matters [31]. Meteorological-normalization and low-cost sensor studies give additional context for air-quality comparison [9, 20, 21, 26, 30, 51]. They do not provide dated, site-level physical-change labels for the public AQS metadata anchors used here.

Bounded gap. Among the reviewed literature, we found no benchmark that jointly fixes the target path, varies donor participation, evaluates selective risk-coverage, and requires structural abstention when the comparison is insufficient. This is a bounded literature statement, not a claim of universal priority. The contribution is the target-fixed evaluation construction, its availability-normalized residual identity, and its audited abstention boundary.

4 Synthetic benchmark construction

4.1 Target-fixed paired worlds

The primary benchmark is a constructed experiment, not a model of $\text{PM}_{2.5}$ transport or monitoring hardware. It generates bounded target-and-donor panels in memory from independent seed streams. Component, target, donor, and date identifiers have no physical-site identity and no relation to AQS data. This deliberate separation lets the experiment isolate one question: what additional scope information is carried by a comparison channel after the target observation has been held fixed?

Each matched pair contains a local arm and a shared arm with identical target arrays, target noise, base panel, and non-scope stress conditions. The arms differ only in the fraction of the injected target schedule received by available donors. At $q = 0$, donor arrays are also identical, creating an observationally indistinguishable negative control. At $q = 1$, all available donors receive the schedule. Intermediate values implement partial sharing after availability-normalized

weights are applied on each retained date.

4.2 Predeclared stress grid

For every component, the complete $5 \times 4 \times 2^5 = 640$ grid crosses nominal donor participation, signal strength, donor mismatch, availability loss, one-donor contamination, nonlinear raw-scale leakage, and bounded noise. The raw-scale field is an adversarial transformation test; it is not a claim about the frequency of such processes in an operational monitoring network. The construction requires exact target identity, complete grid cells, disjoint calibration and evaluation component sets, and no outcome-dependent configuration changes.

4.3 Relation to the AQS audit

The synthetic benchmark supplies constructed scope labels and therefore supports controlled scope-answerability evaluation. The AQS audit in Section 5 serves a different purpose: it measures whether a qualified comparison can be formed in public records and whether the protocol must abstain. It neither supplies labels for the synthetic experiment nor validates a real measurement mechanism.

5 AQS deployment audit data

The AQS audit is included to study deployment boundaries: whether public records support a qualified cross-site comparison and when the audit must abstain. It does not label local-versus-shared scope and does not identify a physical monitoring mechanism.

5.1 Primary public-data corpus

The primary corpus comprises seven annual EPA AirData daily archives for parameter 88101, PM_{2.5} local conditions, spanning 2019–2025 [43, 45]. The snapshot was acquired on 2026-08-30 UTC. A versioned data gate records each official URL, archive checksum, selected CSV member, and retained row count, enabling reconstruction without placing large raw archives or API responses in the repository.

The canonical signal is the reported **Arithmetic Mean** for 24-HR BLK AVG observations. The predeclared cleaning rule keeps finite values with observation percent at least 75, a non-excluded Event Type, and a nonempty Method Code. It then refuses duplicate records with the same state, county, site, POC, and date key. Thus alternative daily summaries cannot be treated as independent observations. The retained primary inventory is a frozen snapshot; it is not a claim that the public source is immutable or complete for all later AQS updates.

The frozen deployment inventory contains 563 persistent anchors, of which 238 pass the distinct-physical-donor eligibility screen before the separate common-input-window check. These counts describe where the legacy observational workflow can be formed; they are not labels for local scope or physical mechanism.

The daily data are public, but raw archives and API responses are not included in this manuscript repository or its paper assets. The display-only case-study renderer verifies archive checksums locally before reading the same source files. That separation supplies a reproducible source chain without placing raw data, credentials, or local cache files under version control.

Table 4: Frozen v0.3.2 data and audit inventory.

Quantity	Count
Canonical daily PM2.5 records	2,424,793
Monitor time series	1,689
Stable pseudo-anchors from Method Code regimes	124
Stable pseudo-anchors from all-monitor regimes	22
Persistent Method Code anchors	563
Anchors with at least one distinct physical donor	394
Anchors with at least three distinct physical donors	238
Complete common-method comparisons	228
Donor-insufficient anchors	325
Estimator input-window failures	10

Note. The physical donor identifier is State Code + County Code + Site Number; at most one POC is retained per donor site.

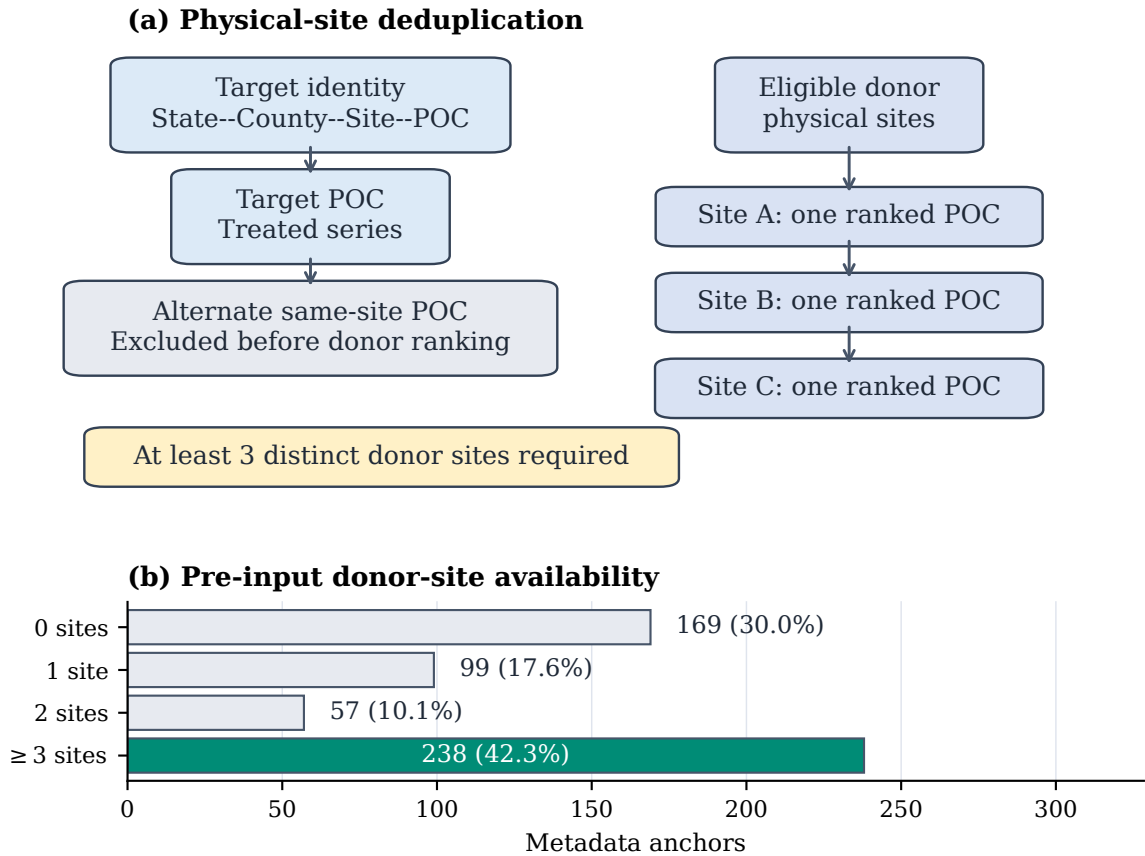


Figure 1: Physical-site donor construction and pre-input availability in the frozen 88101 inventory (563 metadata anchors). A target identity includes State–County–Site–POC; alternate POCs at the target physical site are excluded, and at most one ranked POC is retained at each other physical site. Only 238 of 563 anchors reach the three-distinct-site threshold before the common input-window check.

5.2 Metadata anchors and distinct physical donors

An anchor is a persistent reported Method Code transition with at least 60 calendar days on each side, at least 45 retained observations in each adjacent method run, and no more than a seven-day transition gap. Its fields preserve the old and new reported codes and method names, but neither field establishes why the record changed. EPA network-data-alignment documentation supplies method context but does not date or explain an individual site transition [42]. An eligible geographic donor must be at a different physical site, within the predeclared 100-km radius, method-stable over the target window, observed on at least 60 paired pre-event days, and correlated at least 0.60 on the log scale before the anchor.

Physical donor identity is State Code + County Code + Site Number. A target’s candidate donor list can contain multiple POCs for one physical site, but the ranked donor rule retains at most one: highest pre-event correlation, then shorter distance, then more paired days, then POC as a deterministic tie-breaker. An earlier internal inventory could count multiple POCs from one site as independent geographic donors. Every AQS result reported here uses the corrected physical-site rule; the correction does not affect the separate target-fixed benchmark, which has no AQS inputs. Same-site alternate POCs are reserved for contextual consistency analysis.

5.3 Stable-regime benchmark inputs

Real anchors do not provide known physical-bias truth. The benchmark therefore constructs pseudo-anchors only inside stable target and donor method regimes, at least 60 days away from a reported target or donor transition. The frozen case manifest contains 124 method-transition stable regimes and 22 all-monitor stable regimes. Both sources are retained because the target is a known, stable measurement record at the pseudo-anchor; neither is used as a real-event causal label. Section 7 explains how the target-plus-donor footprints are split before any held-out metric is calculated.

5.4 Audit disposition and parameter separation

Every anchor is retained in the event audit. A complete comparison means that the required common cross-site inputs exist; donor insufficiency and input-window failure are explicit machine-readable outcomes rather than omitted rows. This complete accounting is necessary because a result on only the well-matched subset would overstate the scope of cross-site inference.

Parameter 88502 is processed through a separate data gate, anchor inventory, and audit table. It is neither appended to nor pooled with the primary 88101 results. This separation prevents an apparent replication from being created by mixing parameter-specific reporting conventions, availability, or eligible event populations.

6 Selective audit methods

6.1 Policies for a specified information channel

Section 4 defines paired worlds; the policies here define what an auditor may do with each allowed channel. The target-only forced policy receives a signed target-change score and always predicts local. The comparative forced policy receives a target-minus-availability-normalized-donor residual score and uses a calibration-selected threshold. Four confidence-selective policies use the same comparative prediction with separate, calibration-chosen cutoffs for $\alpha = .01, .05, .10, .20$.

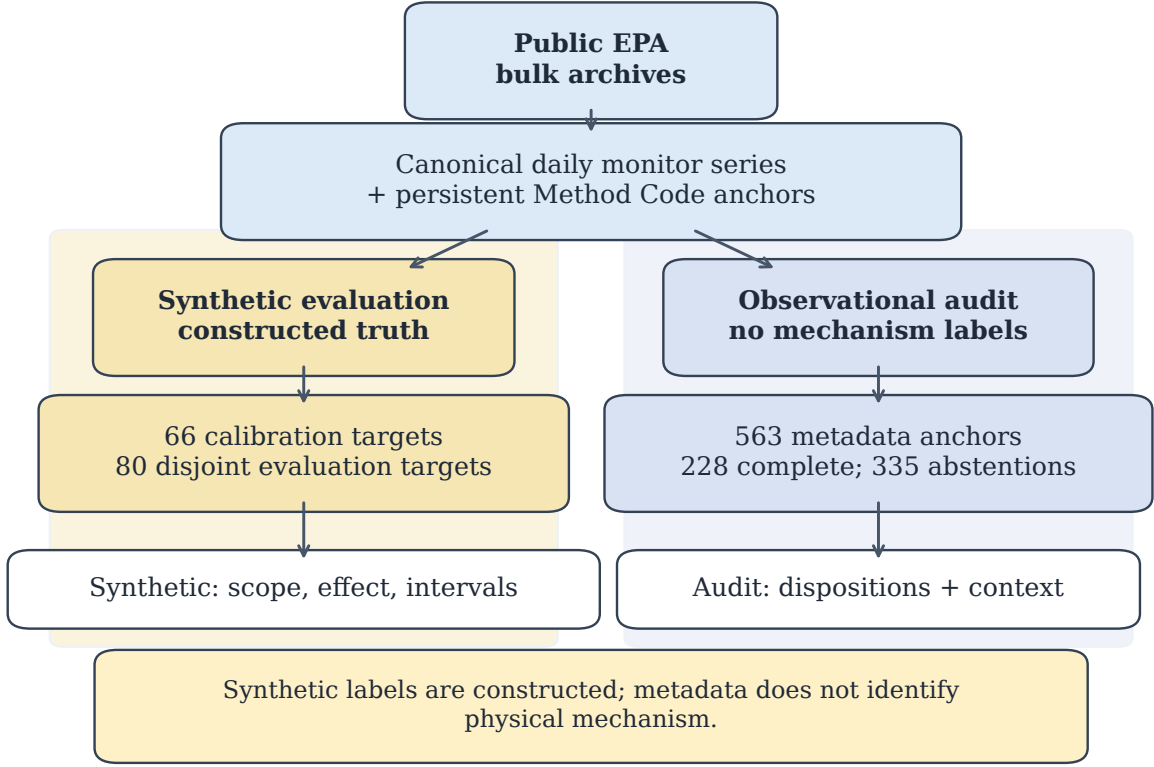


Figure 2: AQS deployment-audit workflow. Shared preprocessing produces canonical daily series and metadata anchors, then feeds a stable-window synthetic lane with constructed labels and an observational lane with audit dispositions. The observational lane supplies diagnostic context, not mechanism labels.

The certificate policy is different: it answers only when the structural margin κ in Section 2 is strictly positive.

6.2 AQS comparison workflow

The independent AQS audit follows a parallel but more limited workflow. Shared preprocessing turns public archives into canonical daily series and metadata anchors. Stable-window synthetic cases have constructed truth; observational anchors instead produce an audit disposition and diagnostic context. This separation prevents the real-data metadata from becoming an unobserved scope label.

6.3 Cross-site comparison methods

For a complete anchor, the donor pool is ranked by pre-event correlation, distance, paired days, and POC after physical-site deduplication. Standard synthetic control solves the nonnegative simplex problem

$$\min_{\substack{w_j \geq 0 \\ \sum_j w_j = 1}} \frac{1}{|T_{\text{cal}}|} \sum_{t \in T_{\text{cal}}} \left(z_{i,t} - \sum_{j \in N_a} w_j z_{j,t} \right)^2.$$

The intercept is represented by calibration centering in the residual definition of Section 2; no post-anchor outcome is supplied to the fit.

Table 5: Method roles in the independent stable-regime benchmark and AQS deployment audit.

Method	Main input and fit	Metadata prior	Role
Standard synthetic control	Target and donor paths; nonnegative pre-anchor fit.	None.	Cross-site reference method in benchmark and AQS audit.
Fixed MetaShift	Target and donor paths; regularized pre-anchor fit.	Correlation–distance reliability prior.	Cross-site comparator in benchmark and AQS audit.
Cross-validated MetaShift	Same fit family with pre-period penalty selection.	Correlation–distance reliability prior.	Benchmark-only comparator; not introduced into the AQS audit after inspection.
Nearest-neighbor DiD	Target and highest-ranked donor.	Donor ranking only.	Cross-site benchmark comparator.
Single-series baselines	Target path alone.	None.	Detection contrast; no qualified regional counterfactual.

Table 6: Two-level donor rule in the AQS deployment audit.

Stage	Requirement	Purpose
Event eligibility	At least three distinct physical donor sites pass the geographic, stability, paired-day, and correlation screens.	Avoids counting multiple POCs from one site as independent geographic evidence.
Date-level calculation	At least two eligible donor sites are observed, with at least 30 retained observations in every required window.	Ensures a numerical residual can be calculated after date-level availability filtering.

Fixed-weight MetaShift begins from the pre-event reliability preference

$$\pi_{ij} \propto \max(\rho_{ij}, 0)^2 \exp(-d_{ij}/50),$$

where ρ_{ij} is pre-event log-scale correlation and d_{ij} is distance in km. Its real-audit objective is

$$\min_{\substack{w_j \geq 0 \\ \sum_j w_j = 1}} \text{MSE}_{T_{\text{cal}}}(z_i, Z_N w) + 0.1 \sum_j w_j^2 + 0.1 \sum_j (w_j - \pi_{ij})^2.$$

In the benchmark, cross-validated MetaShift chooses from a frozen penalty grid using the final 45 pre-event days and refits on the supplied calibration period. It never sees post-anchor data during selection.

6.4 Two-level donor rule

The AQS protocol distinguishes event eligibility from date-level calculation feasibility (Table 6). This preserves the meaning of a cross-site comparison even when a small number of otherwise eligible donors are temporarily missing.

Screening and correlation assembly scale with the available pre-event target–donor pairs, with deterministic donor ranking adding $O(D \log D)$. The retained donor set is capped at three to five sites.

Table 7: Predeclared primary evidence-tier rules and abstention outcomes.

Required diagnostic or condition	Threshold	Audit outcome
Complete common-method comparison	required	Missing input $\Rightarrow$ inconclusive
Selection-aware nested interval	required	Unavailable $\Rightarrow$ inconclusive
Unique stable post-transition time placebos	50	Fewer dates $\Rightarrow$ inconclusive
Raw time-placebo probability	0.10	Above cutoff $\Rightarrow$ not supported
BH-adjusted time-placebo q value	0.10	Unavailable $\Rightarrow$ inconclusive; above cutoff $\Rightarrow$ not supported
Fixed-weight conditional diagnostic interval	exclude 0	Otherwise $\Rightarrow$ not supported
Leave-one-donor-out direction fraction	0.90	Below cutoff $\Rightarrow$ not supported
All available required conditions	pass	Supported candidate discontinuity

Note. The rules synthesize observational diagnostics; they are not a supervised classifier, instrument-fault label, or physical-causality test. The nested interval is required for tier assignment but is not claimed to have synthetic coverage calibration.

6.5 Intervals, falsification, and audit dispositions

For fixed donor weights, the primary interval resamples pre- and post-anchor residuals using circular moving blocks of seven days and 1,000 fixed-seed repetitions. It is conditional on previously chosen donors and weights. The selection-aware nested procedure re-block-resamples the calibration matrix, recomputes eligibility, selects donors from the fixed candidate pool, refits weights, and resamples the pre/post residual windows. Both are diagnostic intervals in the real audit, not calibrated estimates of physical bias.

The audit also computes stable post-transition time placebos, donor-as-treated placebos, a date-resampling comparison, and leave-one-donor-out refits. These are falsification and robustness diagnostics; they neither create random assignment nor identify a physical mechanism.

An event is inconclusive if a required comparative or diagnostic input is unavailable. It is not supported when all required diagnostics are available but at least one fails. A supported candidate satisfies every required diagnostic. Benjamini–Hochberg adjustment is only an exploratory multiple-screening quantity [12]; the resulting tiers are observational audit summaries rather than labels for instrument failure, replacement, bias, or pollution correction.

7 Experimental design

7.1 Primary target-fixed evaluation

The primary answerability experiment was implemented and protocol-checked before evaluation outputs were created. It uses bounded synthetic panels only; AQS data, AQS metadata, earlier evidence releases, and taxonomy decisions are not inputs. Each component has 300 dates, one target, four donors, a fixed anchor day, equal donor weights, and independently generated bounded common and idiosyncratic innovations.

Calibration contains 120 components and evaluation contains 360 disjoint components. The complete design has 76,800 calibration pairs and 230,400 evaluation pairs. Each component contributes 640 matched pairs, and each pair contains two scope arms:

Sample accounting. 120 calibration components $\rightarrow$ 76,800 pairs; 360 held-out components $\rightarrow$ 230,400 pairs $\rightarrow$ 460,800 scope arms. Uncertainty is resampled at the component level because pairs and arms within a component share its generated panel.

The $5 \times 4 \times 2^5 = 640$ grid crosses nominal participation $q \in \{0, .25, .50, .75, 1\}$, signal strength $H \in \{.04, .08, .12, .20\}$, donor mismatch, availability loss, one-donor contamination, a common

nonlinear raw-scale field, and bounded noise. Every cell produces one local and one shared arm. The target arrays must be exactly identical within a pair; at $q = 0$, donor arrays must be identical as well. The protocol fails closed on target-identity violations, footprint leakage, stale hashes, incomplete cells, calibration/evaluation mixing, or outcome-dependent configuration changes.

7.2 Policies, thresholds, and uncertainty

A calibration-only threshold selects the comparative forced policy. Confidence cutoffs are selected from 101 calibration quantiles separately for each predeclared tolerance. Evaluation reports every policy at every reporting tolerance; it does not select a new policy or retune a cutoff after outcomes. The certificate threshold is analytic and uses the declared bounds plus realized retained-date availability and panel values required by the clipping-aware raw-field bound. The ambiguous partial-scope region consists of all $q > 0$ pairs with nonpositive structural margin; the certificate must abstain there. The $q = 0$ stratum is retained separately as observationally unanswerable.

The unit of uncertainty is the evaluation component rather than a scope arm, date, or grid row. A 1,000-repetition percentile component bootstrap with a prespecified seed reports policy risk-coverage, finite-policy envelopes, certificate behavior, and subgroups without collapsing them into one favorable score.

7.3 Independent stable-regime benchmark

The AQS-based stable-regime benchmark supplies a separate, known-effect comparison of cross-site and single-series methods. It contains 146 physical target sites: 66 for calibration and 80 for held-out evaluation. Whole connected components of the complete target-plus-donor physical-input graph are assigned to one split, so no physical input site crosses calibration and evaluation.

The benchmark injects target-only additive, proportional, gradual-drift, temporary-step, and variance changes, together with matched target-and-donor regional variants. Each variant has 400 held-out samples. Ranking thresholds are selected only on calibration cases. Local-effect MAE, AUPRC, macro-F1, precision, recall, and regional false-positive rate are reported as complementary metrics; the variance-only condition has no defined level-effect truth and therefore no MAE.

7.4 AQS deployment audit and interval protocol

The AQS audit applies the distinct-physical-donor construction, pre-event fitting boundaries, and effect windows to all 563 metadata anchors. Cross-validated MetaShift is benchmark-only and is not introduced into the real audit after its outcomes are known. Predeclared sensitivity analyses vary one factor at a time and record availability and interpretation dependence, not a search for a favorable setting.

Fixed-weight interval coverage is evaluated on held-out synthetic effects with fixed confidence level, seven-day blocks, perturbations, and 1,000 repetitions. The prespecified full nested protocol exceeded the fixed computational budget and was not executed; consequently, no selection-aware coverage claim is made. In the real audit, the calibration slice $t_0 - 180$ through $t_0 - 15$ overlaps the displayed pre-anchor slice $t_0 - 60$ through $t_0 - 1$ by 46 days. No post-anchor data enter the fit, but the displayed pre-anchor residuals are not an independent validation sample.

Same-site POC records, narrow hourly windows, QA availability, and targeted official-document review provide contextual checks with known limits. A separate parameter-88502

pipeline uses its own scan and audit; it is never pooled with the primary 88101 results.

8 Results

8.1 RQ1: Comparative information expands answerability only at a relaxed tolerance

Answer. On the target-fixed held-out grid, the target-only forced policy answered every scope arm but had 50% observed conditional scope error. Thus no predeclared target-only policy had positive finite-policy coverage at any reported tolerance below 0.5. The comparative forced policy also answered every arm but had 27.5% observed error. Its confidence-selective policies calibrated at .01, .05, and .10 answered 0 held-out arms. Only the policy calibrated at .20 reached a positive comparative envelope: 60.4% coverage at 19.5% observed conditional error.

Table 8: Frozen finite-policy scope-answerability envelope on the held-out target-fixed evaluation components. Parentheses give observed conditional scope error for the policy attaining the displayed coverage; – means no positive-coverage candidate qualified. The certificate-assisted channel uses synthetic design information and is not an operational deployment channel. Its 0 observed errors are counted among 179,994 answered scope arms, not estimated population risk.

Tolerance α	Target-only	Comparative	Certificate-assisted
1%	0%	0% (–)	39.1% (0 observed errors)
5%	0%	0% (–)	39.1% (0 observed errors)
10%	0%	0% (–)	39.1% (0 observed errors)
20%	0%	60.4% (19.5%)	39.1% (0 observed errors)

At $\alpha = .20$, the achieved finite-policy Scope Answerability Gain from the target-only to comparative channel is 60.4%. Its component-bootstrap interval is [60.1%, 60.7%]. This is a comparison among frozen policies on held-out synthetic components, not an estimate of a universal optimum or future population risk.

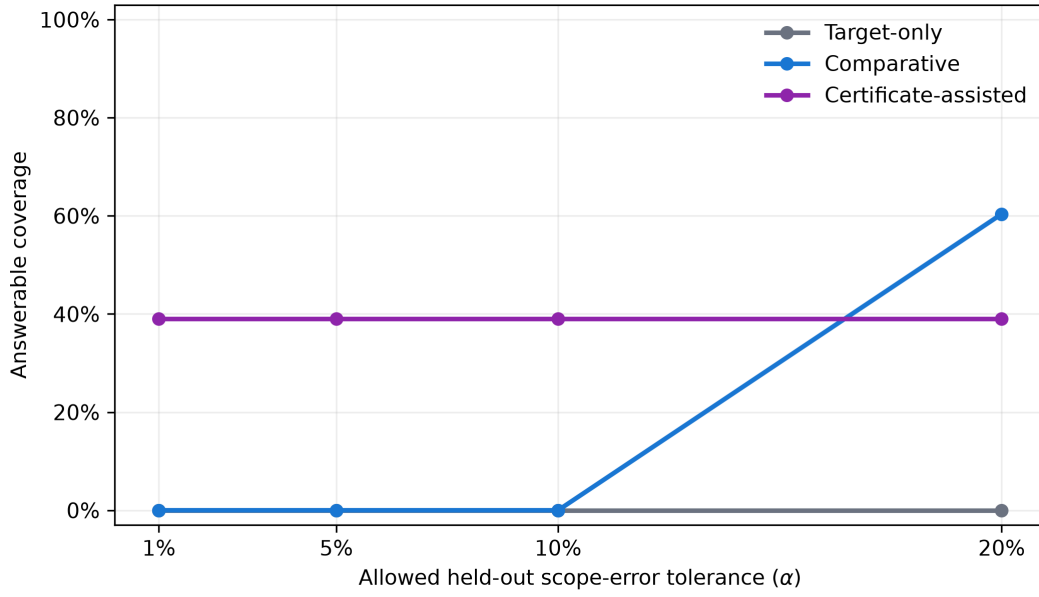


Figure 3: Held-out finite-policy answerability frontiers by information channel. Target-only policies do not reach positive coverage at the four reported tolerances. Comparative coverage emerges only at $\alpha = .20$. The certificate-assisted policy is plotted separately because it uses bounded synthetic-design information, not a real-network deployment channel.

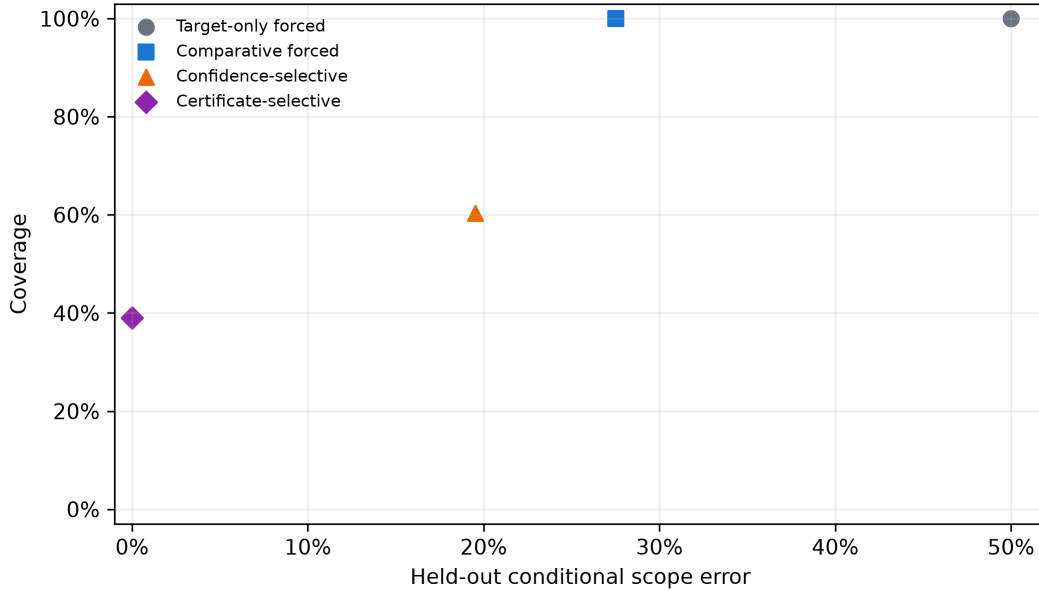


Figure 4: Risk-coverage positions of the predeclared held-out policies. Forced comparative classification has full coverage but high error; strict confidence policies abstain completely. The certificate has lower coverage than forced comparison but no observed errors under its bounded construction.

8.2 RQ2: The structural certificate delineates a retained abstention boundary

Answer. The structural interval-separation certificate answered 89,997 of 230,400 matched pairs (39.1%), with 0 observed errors among 179,994 answered scope arms. It recovered 97.7% of pairs in the predeclared simulation-information oracle region and abstained on the remaining 140,403 pairs (60.9%), including all 46,080 observationally indistinguishable $q = 0$ negative controls. It had 0 observed envelope violations. These finite-grid observations hold only for the bounded generator: they neither establish zero population risk nor yield a deployable real-network certificate.

Table 9: Certificate behavior by nominal donor participation on held-out target-fixed pairs. Oracle-region recovery is the share of predeclared simulation-information-oracle answerable pairs recovered by the certificate; it is undefined at $q = 0$ because that negative-control stratum has no oracle-answerable pairs. Across these strata, 0 observed errors means 0 of 179,994 answered scope arms, not population risk.

Participation	Pair coverage	Observed error	Oracle-region recovery
$q = 0\%$	0%	–	–
$q = 25\%$	10.9%	0 observed errors	87.5%
$q = 50\%$	38.3%	0 observed errors	94.2%
$q = 75\%$	66.4%	0 observed errors	98.8%
$q = 100\%$	79.7%	0 observed errors	100%

The certificate coverage increases with nominal donor participation and signal, but it is intentionally incomplete under contamination, nonlinear raw fields, and heteroskedastic noise. In particular, the $q = 0$ stratum contains 46,080 matched pairs with identical target and donor observations in both arms, produces 0 certificate answers, and has no oracle-answerable pair. The result does not declare all real partial scope objectively ambiguous; it marks where this sufficient bound does not authorize an answer.

Table 10: Predeclared failure and abstention accounting on 230,400 held-out target-fixed matched pairs. These rows are retained rather than converted into a favorable single score. The rows overlap and must not be summed: the $q = 0$ row is a subset of the nonpositive-margin row, while arm-level error rows use a different denominator from pair-level rows.

Outcome	Count	Interpretation
Target and donor observations identical at $q = 0$	46,080	negative-control abstention stratum
Nonpositive structural margin	140,403	certificate must abstain
Certificate-answered pairs	89,997	0 errors among 179,994 answered arms
Envelope violations	0	none observed
Comparative-forced errors	126,764	among 460,800 forced scope arms
Confidence policy at calibration $\alpha = 0.05$	0	no answered arms

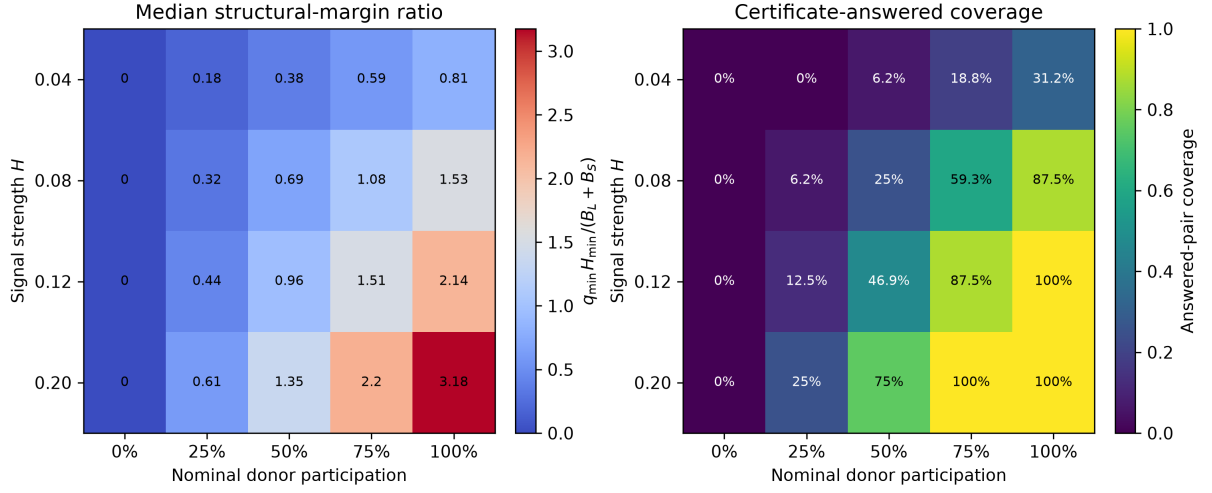


Figure 5: Normalized structural-margin phase diagram from the held-out grid. The left panel shows median $q_{\min} H_{\min} / (B_L + B_S)$; values above one meet the sufficient interval-separation threshold. The right panel shows certificate answered-pair coverage. Both panels average over the five adverse comparison factors and describe the bounded synthetic construction only.

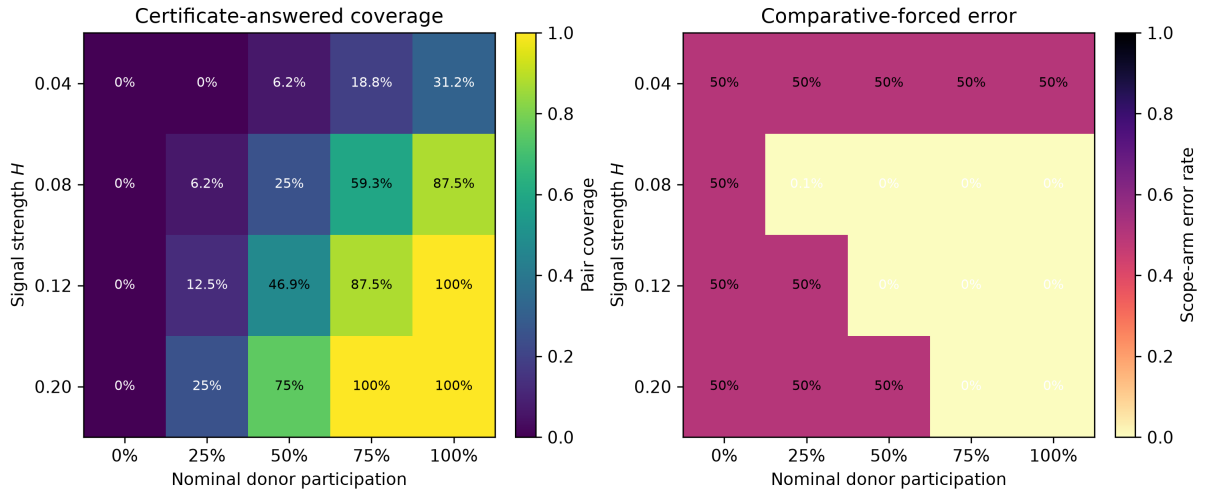


Figure 6: Retained failure map across signal and nominal donor participation. The left panel gives certificate coverage; the right panel gives comparative-forced error. Low signal and adverse comparison conditions remain in the evaluation rather than being filtered away. The $q = 0$ column is an observational-identity negative control.

8.3 RQ3: Cross-site comparison helps, but MetaShift has no supported aggregate advantage

Answer. In the independent stable-regime benchmark, all four cross-site methods have aggregate AUPRC between 0.884 and 0.915, whereas the five target-only methods are near 0.50 (Table 17 in Appendix A.1). This illustrates the value of comparative information for the constructed local-versus-regional task, rather than a generic change-detection claim.

Table 11: Independent stable-regime synthetic evaluation.

Method	MAE	AUPRC	Macro-F1	Regional FPR
Standard SC	0.100	0.915	0.816	0.140
MetaShift fixed	0.103	0.898	0.795	0.150
MetaShift CV	0.097	0.902	0.809	0.135
Nearest-neighbor DiD	0.111	0.884	0.789	0.075

Note. Metrics are aggregates over the frozen `stable_full_v2` evaluation set. Lower MAE and regional false-positive rate are preferable; higher AUPRC and macro-F1 are preferable.

Standard synthetic control has the strongest aggregate AUPRC and macro-F1. Cross-validated MetaShift has the smallest aggregate local-effect MAE, while nearest-neighbor difference-in-differences has the lowest regional false-positive rate. No method wins every metric. Most importantly, neither MetaShift variant has a confidence-supported aggregate MAE advantage over standard synthetic control: both paired component-bootstrap intervals include zero (Table 12).

Table 12: Paired event-cluster bootstrap for local-effect MAE differences.

Comparison	Difference	95% bootstrap interval	Clusters
MetaShift fixed minus Standard SC	0.004	[-0.004, 0.012]	80
MetaShift CV minus Standard SC	-0.002	[-0.008, 0.003]	80

Note. Difference is MetaShift minus standard synthetic control; negative values favor MetaShift on MAE. Both bootstrap intervals include zero.

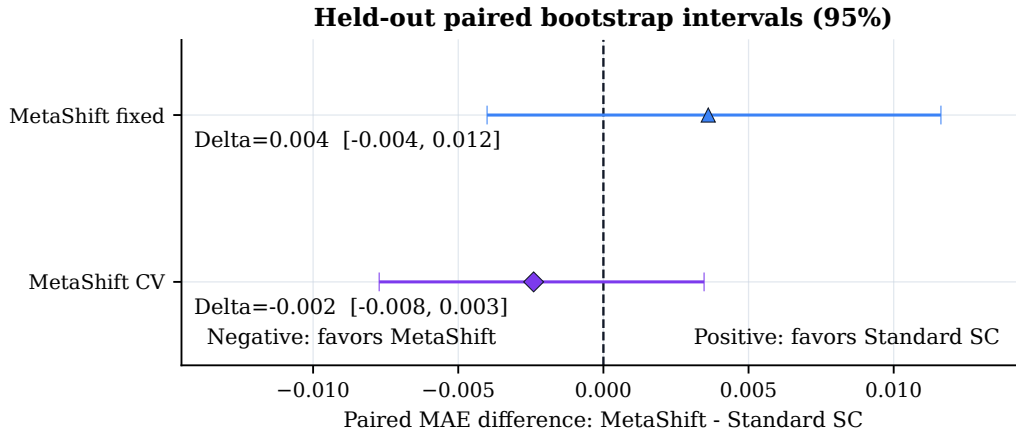


Figure 7: Paired event-cluster bootstrap intervals for MetaShift minus standard synthetic-control local-effect MAE. Negative values favor MetaShift; both intervals cross zero. The frozen evidence therefore does not support an aggregate MetaShift-superiority claim.

8.4 RQ4: A qualified AQS comparison is available for 228 of 563 anchors

Answer. The distinct-physical-donor rule permits a complete common cross-site comparison for 228 of 563 anchors (40.5%). The other 325 donor-insufficient and 10 input-window failures are retained as abstentions, not dropped observations. This availability result is the AQS audit’s primary deployment finding.

Table 13: Complete 88101 event audit and real-event diagnostics.

Audit disposition		Anchors	
Complete common-method comparison		228	
Fewer than three distinct physical donors		325	
Estimator input-window failure		10	
Total		563	

Method	Events	Fixed conditional interval excludes 0	Median log effect
MetaShift fixed	228	159	-0.087
Standard SC	228	145	-0.071
Nearest-neighbor DiD	228	134	-0.075

Note. Nested selection-aware intervals are available for 227/228 complete events; 153 exclude zero. Their mean width is 0.191 log units versus 0.179 for fixed-weight MetaShift intervals. Leave-one-donor-out refits complete for 227 events, with 202 retaining direction under every donor removal. All intervals are diagnostic, not calibrated physical-bias confidence intervals.

The exploratory evidence synthesis assigns 34 supported candidates, 122 not-supported events, and 407 inconclusive events. “Supported” means that all predeclared diagnostic conditions were available and passed; it does not confirm a physical failure, replacement, measurement bias, or pollution correction.

(a) Availability partition --- all 563 anchors**(b) Evidence-tier partition --- 228 complete comparisons**

Overall disposition: 34 supported candidates | 122 not supported | 407 inconclusive
 407 = 325 donor-insufficient + 10 input-window failures + 72 complete but inconclusive

Figure 8: AQS audit accounting. Panel (a) partitions all 563 metadata anchors into donor-insufficient, input-window-failure, and complete comparison dispositions. Panel (b) partitions complete comparisons into protocol evidence tiers. These are audit dispositions, not diagnoses of physical monitor changes.

8.5 RQ5: Calibration and contextual checks limit, rather than validate, real-data interpretation

Answer. The fixed-protocol interval constructions do not support a generic calibration claim. On the held-out synthetic effects, nominal 95% fixed-weight coverage ranges from 63.9% to 67.3% over 3,200 instances per method, while nominal 90% split-conformal coverage ranges from 98.8% to 99.6% and is conservative.

Table 14: Held-out synthetic interval coverage by method.

Method	Fixed 95% coverage	Fixed mean width	Conformal 90% coverage	Conformal mean width
Standard SC	67.3%	0.265	98.8%	0.396
MetaShift fixed	63.9%	0.270	99.4%	0.471
MetaShift CV	65.4%	0.265	98.9%	0.437
Nearest-neighbor DiD	66.7%	0.287	99.6%	0.565

Note. Each method has 3,200 evaluation effect instances. Fixed-weight intervals under-cover and split-conformal intervals over-cover under this frozen synthetic design.

The real-data date-resampling diagnostic holds each event’s pre-transition donor weights fixed and compares the mean standardized score at the metadata anchor with one randomly selected stable post-transition date per eligible event. Under the within-event exchangeability diagnostic null, its statistic is the difference between the anchor mean and sampled-placebo mean; the reported one-sided tail probability is $(1 + \#\{\text{sampled placebo mean} \geq \text{anchor mean}\})/201 = 0.00498$. It is a comparison with stable dates, not a causal probability or a physical-mechanism test.

Of 11 same-site POC candidates, 9 have usable paired hourly windows and 8 agree in daily/hourly direction. Targeted official-document review finds 0/20 dated site-specific confirmations. Missing documentation and unavailable POC data remain unavailable evidence; they are not evidence that nothing changed.

Table 15: Placebo, external-document, and same-site POC evidence boundaries.

Diagnostic	Value
Complete time-placebo events with at least 50 dates	157
Complete time-placebo events with 100 dates	128
Raw time-placebo probabilities at most 0.10	71
BH q values at most 0.10	40
Donor-as-treated placebo records	866
Date-resampling permutations	200
Dated site-specific external confirmations	0/20
Usable hourly same-site POC comparisons	9/11
Daily/hourly POC direction agreements	8/9
Standard SC 90th-percentile risk-gate retention	75/80

Note. Time and donor placebos are diagnostic falsifications. Same-site POC and document review provide contextual evidence only, not instrument ground truth.

9 Discussion

9.1 Main answer

The central result is conditional rather than triumphant: a target-only detector can recognize a change, yet cannot resolve its local-versus-shared scope when matched target observations carry no scope information. Comparative information enlarges the achieved held-out answerability frontier only at the least strict reported tolerance. The experiment therefore rejects both extremes: target information alone is insufficient in the stipulated paired worlds, and donor comparison does not automatically make every scope question answerable.

9.2 Why comparison sometimes fails

The partial-scope grid makes the failure boundary visible. Low donor participation, weak target signal, missing donors, contamination, nonlinear raw-scale leakage, and noise shrink the interval gap on which the certificate relies. A nonpositive structural margin is not treated as a low-confidence label: the certificate abstains. This is useful precisely because the failed and unanswerable cells remain in the result rather than being discarded to improve a classifier score.

9.3 Why estimator complexity did not win

The stable-regime benchmark confirms that cross-site information is useful for the constructed scope task, but it does not support a general MetaShift advantage. Standard synthetic control leads aggregate ranking and macro-F1; cross-validated MetaShift has the best point estimate for local-effect MAE, but its paired uncertainty interval versus standard synthetic control crosses zero. The appropriate contribution is consequently the information-and-abstention framework, not a claim that a more complex estimator is universally better.

9.4 Implications for real audits

The AQS inventory measures whether a qualified comparison can be formed, not whether a monitor physically changed. Its many donor-insufficient and input-window-failure dispositions quantify where this audit cannot answer. A supported candidate only prioritizes an anchor for technical-record review. Station histories, maintenance records, and human domain review remain necessary for a mechanism claim; neither metadata anchors, residuals, placebos, nor same-site POC agreement substitute for them.

Table 16: Assumptions and claim boundaries for the target-fixed scope-answerability experiment.

Assumption	Supports	Does not support
Balanced labels, matched target laws, and no target-side leakage	Target-only impossibility in the stipulated paired worlds.	Target-only impossibility for arbitrary priors, leaked features, or AQS records.
Nested target-only and comparative channels under one law	Policy-set monotonicity under information refinement.	Improvement by every finite-sample classifier.
Analysis-scale intervention, fixed weights, and arm-invariant retained dates	The exact construction-specific residual identity.	Exact raw-scale algebra after non-linear transformation or clipping.
Declared bounded mismatch, noise, leakage, and contamination envelopes	Interval separation when $\kappa > 0$.	Unbounded-noise, population-risk, or real-network guarantees.
Known synthetic participation and bounds	Certificate coverage and oracle-region recovery.	An operational external-evidence policy.

10 Limitations

10.1 Scope of the answerability claims

Table 16 separates the assumptions supporting the target-fixed answerability statements from claims they do not support. The certificate relies on declared bounded generator quantities, including participation, mismatch, leakage, and noise envelopes. It is therefore a synthetic-design-assisted sufficient condition, not a confidence guarantee, an empirical O_E evidence channel, or a deployable sensor-network rule.

10.2 Limits of the AQS audit

AQS Method Code transitions are metadata anchors, not physical ground truth. Pre-event matching, distinct physical donors, time and donor placebos, and leave-one-donor-out checks constrain obvious comparison failures, but do not identify a unique cause of a residual. The $t_0 - 180$ through $t_0 - 15$ calibration slice overlaps the displayed $t_0 - 60$ through $t_0 - 1$ pre-anchor slice by 46 days. No post-anchor outcome enters fitting, but the displayed pre-anchor residuals are consequently not an independent validation sample.

Fixed-weight moving-block intervals under-cover known synthetic effects, and split-conformal intervals over-cover them. The unexecuted full selection-aware synthetic coverage study was outside the fixed computational budget. Thus real-event intervals remain diagnostic descriptions, not coverage-calibrated confidence intervals for physical measurement bias.

10.3 Transfer and human-review requirements

The archive years, PM_{2.5} parameter, U.S. network, availability rules, donor radius, and synthetic intervention families constrain transfer. The secondary 88502 pipeline demonstrates only sparse separate-pipeline feasibility. A physical mechanism, equipment diagnosis, or causal measurement-bias claim would require dated station histories, calibration and maintenance records, independently reviewed taxonomy, and human domain judgment. The complete threats-to-validity table and applicability map are retained in the appendix rather than omitted. No taxonomy-stratified analysis is reported.

11 Conclusion

MetaShift-Bench studies a question that ordinary change detection leaves open: when can an observed change be called local rather than shared? Its target-fixed paired design turns that question into a selective prediction problem. At a stated tolerated scope error, the answerability frontier measures how much of the stipulated task a channel can answer; the structural certificate abstains where bounded comparison evidence cannot separate the two scopes.

The held-out findings are deliberately mixed. Target-only scope inference has no positive finite-policy coverage below one-half error in the matched paired construction. Comparative information gains coverage only at the least strict reported tolerance. The certificate has no observed errors where its declared intervals separate, but abstains on most pairs and is not a real-network deployment policy. The separate AQS audit records comparison availability and diagnostic abstentions, while preserving the negative result that MetaShift does not have a confidence-supported aggregate advantage over standard synthetic control.

These results support a bounded principle: an audit should state what information its scope decision requires and should abstain when that information is insufficient. They do not establish real monitor mechanisms, population-risk guarantees, physical measurement bias, or corrected concentrations. Those questions require independently documented records and human review.

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A Supplementary evidence and protocols

A.1 Complete method comparisons

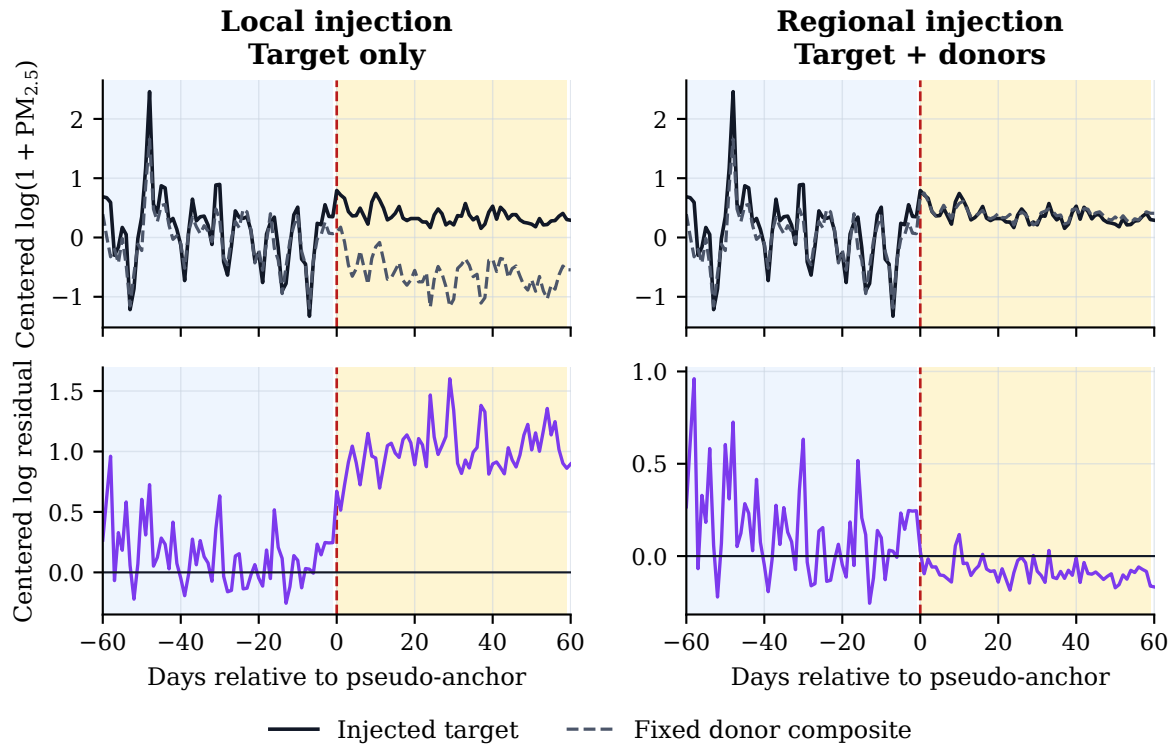
Table 17 restores all nine frozen stable-regime benchmark methods. The five target-only methods remain near chance on the constructed local-versus-regional ranking task; this does not mean they are unusable for ordinary single-series change detection. N/A local-effect MAE means that a method does not estimate the level-effect target for that perturbation design.

Table 17: Complete frozen aggregate comparison of all benchmark methods.

Method	Local-effect MAE	AUPRC	Macro-F1	Regional FPR
Standard SC	0.09983	0.91476	0.81641	0.140
MetaShift fixed	0.10344	0.89832	0.79515	0.150
MetaShift CV	0.09742	0.90178	0.80916	0.135
Nearest-neighbor DiD	0.11150	0.88443	0.78852	0.075
Bayesian mean shift	0.24672	0.50116	0.49962	0.542
Before-after median	0.25330	0.50125	0.50024	0.505
CUSUM	N/A	0.50117	0.49720	0.581
Rolling MAD	N/A	0.50112	0.49427	0.402
PELT	N/A	0.50025	0.49976	0.522

Note. All values are aggregates over the held-out stable-regime evaluation partition. Lower MAE and regional FPR are preferable; higher AUPRC and macro-F1 are preferable. N/A means that the method supplies no local-effect magnitude estimate for that perturbation design, not that the method was omitted.

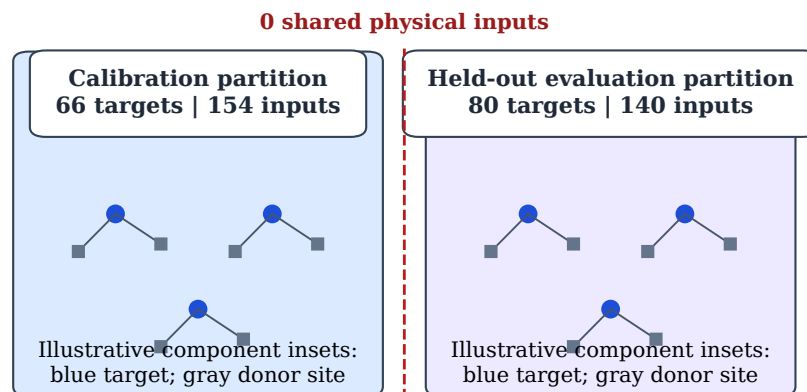
Data-derived stable-window illustration (lexicographically first held-out case)



Blue: 60-day pre window. Amber: 60-day post window. Frozen additive magnitude = 11.86 $\mu\text{g}/\text{m}^3$.

Figure 9: A deterministic stable-regime synthetic illustration. The injected effect and paired local/regional scope labels are known by construction; the underlying AQS paths do not supply physical mechanism labels.

Whole target-plus-donor components are assigned before held-out metrics



Whole components prevent any target or donor physical site from crossing the split.

Figure 10: Connected-component split integrity for the stable-regime benchmark. No physical input site is shared between calibration and held-out evaluation.

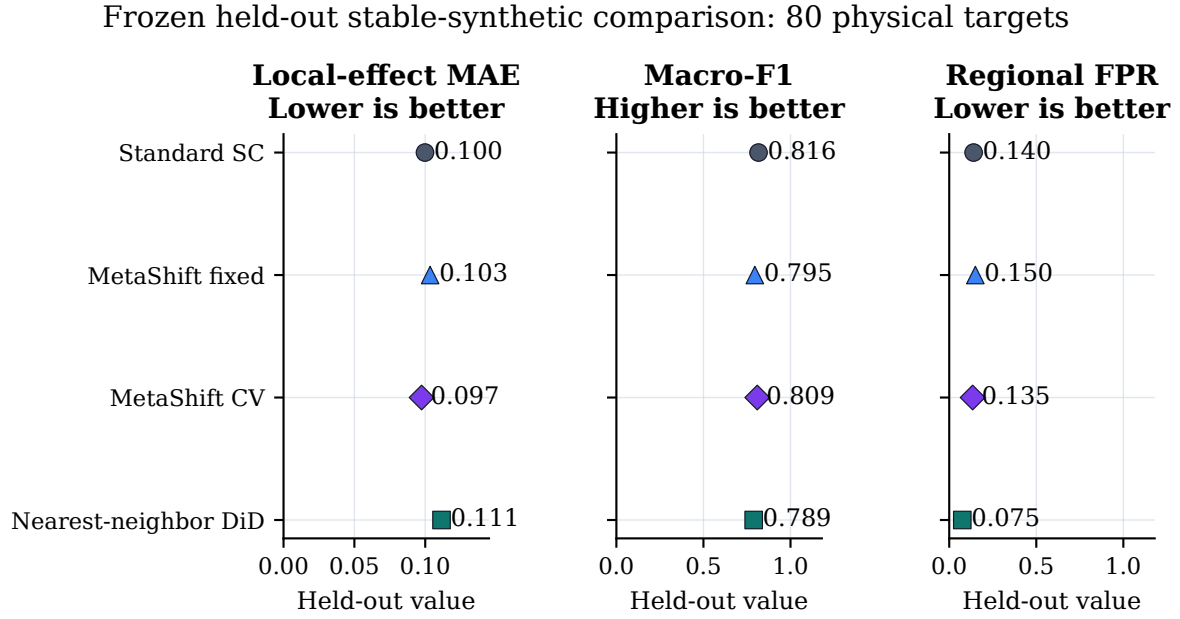


Figure 11: Held-out stable-regime comparison for the four cross-site methods. Dots show local-effect MAE, macro-F1, and regional false-positive rate. These panels do not form a composite ranking.

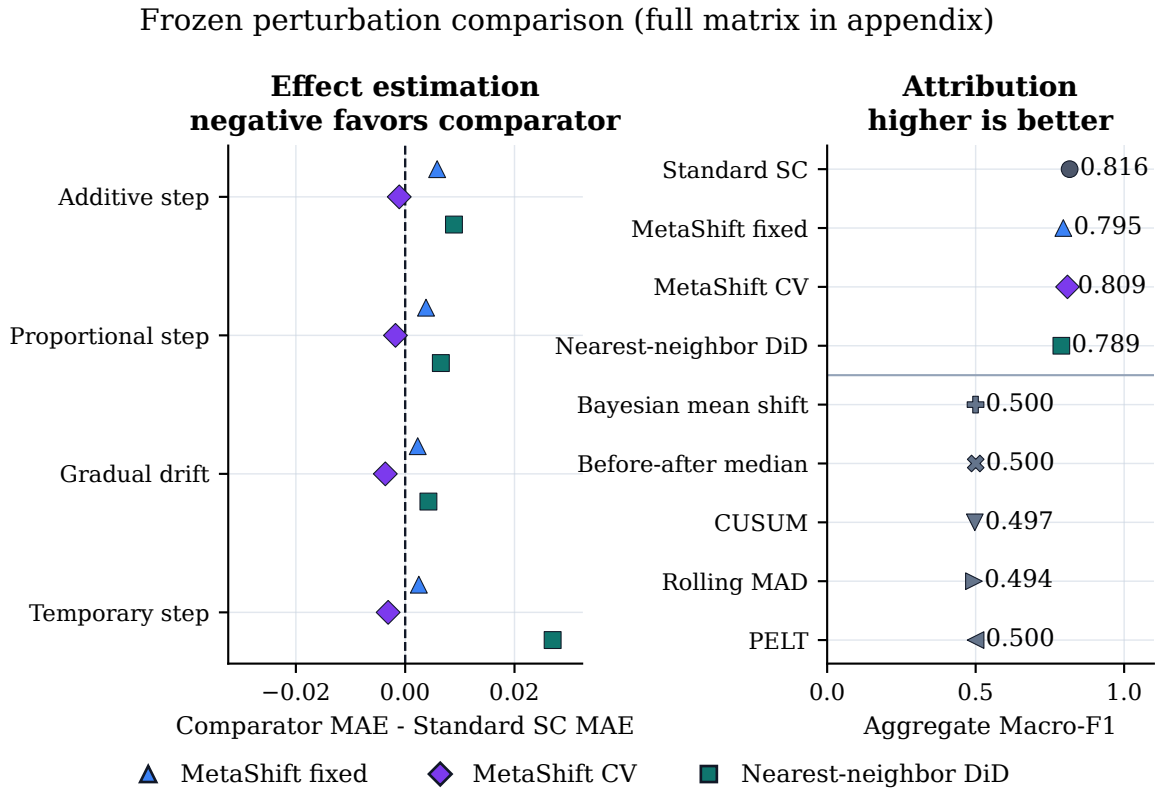


Figure 12: Perturbation-family comparison. The left panel gives local-effect MAE differences from standard synthetic control for cross-site comparators; the right panel gives aggregate macro-F1 for all nine methods. Variance-only local-effect MAE is undefined by design.

Table 18: Perturbation-family-specific held-out synthetic metrics for all methods.

Paired perturbation family	Method	MAE	AUPRC	Macro-F1	Regional FPR
Additive step	Standard SC	0.07788	0.99946	0.93092	0.138
Additive step	MetaShift fixed	0.08373	0.99695	0.87969	0.233
Additive step	MetaShift CV	0.07680	0.99801	0.90161	0.193
Additive step	Nearest-neighbor DiD	0.08680	0.99210	0.93614	0.105
Additive step	Bayesian mean shift	0.22860	0.50497	0.41031	0.890
Additive step	Before-after median	0.22774	0.50497	0.44778	0.808
Additive step	CUSUM	N/A	0.50497	0.43785	0.833
Additive step	Rolling MAD	N/A	0.50497	0.49862	0.553
Additive step	PELT	N/A	0.50497	0.42673	0.858
Proportional step	Standard SC	0.07635	0.85856	0.76319	0.188
Proportional step	MetaShift fixed	0.08016	0.84617	0.78293	0.090
Proportional step	MetaShift CV	0.07457	0.84296	0.78357	0.103
Proportional step	Nearest-neighbor DiD	0.08287	0.82011	0.70938	0.065
Proportional step	Bayesian mean shift	0.23444	0.50497	0.48849	0.350
Proportional step	Before-after median	0.22774	0.50497	0.46944	0.260
Proportional step	CUSUM	N/A	0.50497	0.48467	0.328
Proportional step	Rolling MAD	N/A	0.50497	0.40894	0.108
Proportional step	PELT	N/A	0.50497	0.48326	0.320
Gradual drift	Standard SC	0.08321	0.99710	0.93985	0.110
Gradual drift	MetaShift fixed	0.08551	0.99224	0.89419	0.193
Gradual drift	MetaShift CV	0.07958	0.99521	0.92082	0.153
Gradual drift	Nearest-neighbor DiD	0.08746	0.98570	0.93248	0.085
Gradual drift	Bayesian mean shift	0.24325	0.50497	0.45055	0.800
Gradual drift	Before-after median	0.23075	0.50497	0.47143	0.733
Gradual drift	CUSUM	N/A	0.50497	0.45322	0.793
Gradual drift	Rolling MAD	N/A	0.50497	0.49969	0.475
Gradual drift	PELT	N/A	0.50497	0.49992	0.513
Temporary step	Standard SC	0.16187	0.98113	0.91246	0.108
Temporary step	MetaShift fixed	0.16436	0.96551	0.88871	0.130
Temporary step	MetaShift CV	0.15875	0.96994	0.90372	0.113
Temporary step	Nearest-neighbor DiD	0.18885	0.93733	0.84114	0.065
Temporary step	Bayesian mean shift	0.28058	0.50497	0.49930	0.463
Temporary step	Before-after median	0.32696	0.50497	0.49614	0.588
Temporary step	CUSUM	N/A	0.50497	0.48686	0.660
Temporary step	Rolling MAD	N/A	0.50497	0.45747	0.780

Note. Each family aggregates its target-only local perturbation and its matched target-and-donor regional variant. Consequently, regional FPR is assessed within the same paired family; the rows are not local-only effect estimates. N/A has the same meaning as in Table 17.

Table 18, continued

Paired perturbation family	Method	MAE	AUPRC	Macro-F1	Regional FPR
Temporary step	PELT	N/A	0.50497	0.47750	0.708
Variance increase	Standard SC	N/A	0.58680	0.48825	0.158
Variance increase	MetaShift fixed	N/A	0.57484	0.45372	0.105
Variance increase	MetaShift CV	N/A	0.57314	0.46902	0.113
Variance increase	Nearest-neighbor DiD	N/A	0.56157	0.42775	0.055
Variance increase	Bayesian mean shift	N/A	0.50056	0.45595	0.205
Variance increase	Before-after median	N/A	0.50741	0.42520	0.135
Variance increase	CUSUM	N/A	0.50470	0.48012	0.290
Variance increase	Rolling MAD	N/A	0.50191	0.39371	0.093
Variance increase	PELT	N/A	0.49172	0.45495	0.213

Note. Each family aggregates its target-only local perturbation and its matched target-and-donor regional variant. Consequently, regional FPR is assessed within the same paired family; the rows are not local-only effect estimates. N/A has the same meaning as in Table 17.

Table 19: Reliability-prior and regularization ablations on shared synthetic inputs.

Variant	MAE	Macro-F1	Regional FPR
Standard SC	0.09983	0.81641	0.140
No ridge penalty	0.10289	0.80127	0.138
Ridge = 0.01	0.10304	0.80263	0.130
Add coverage	0.10332	0.79862	0.135
MetaShift full prior	0.10344	0.79515	0.150
No reliability-prior penalty	0.10350	0.80167	0.133
No correlation term	0.10362	0.79858	0.133
No distance term	0.10370	0.80588	0.088
Uniform prior	0.10378	0.80521	0.112
Ridge = 1.0	0.10500	0.80285	0.097
Direct reliability	0.10543	0.80574	0.105

Note. The common standard-synthetic-control records align exactly across the main and ablation experiments to tolerance 10^{-10} .

The reliability-prior ablations share the standard-control rows exactly to the configured numerical tolerance. They do not establish a stable MetaShift advantage, but make the frozen design variants available for inspection.

A.2 Historical endpoint and supplementary answerability diagnostics

The immutable v0.4 construction is retained only as an endpoint sanity check. When its local and regional worlds hold the target sequence fixed, target-only classification has observed error 0.5, and its constructed comparative rule has zero observed error on that finite endpoint evaluation. It does not measure a risk-coverage frontier, calibration-selected abstention, partial scope, a failure boundary, or real measurement mechanisms.

Figure 13 gives the full frozen certificate-validity summary. Its reader-facing zero-error labels are observed finite-grid values, not population-risk estimates.

A.3 AQS audit protocol

For each primary-parameter anchor $a = (s, p, t_0, m^-, m^+)$, the audit:

1. validates canonical daily observations and identifies a persistent reported Method Code transition;
2. constructs a geographic candidate pool at distinct physical sites, retaining at most one POC per site by the pre-event ranking rule;
3. records donor insufficiency if fewer than three qualified physical donors exist;
4. fits cross-site weights and the residual center only on the inclusive $t_0 - 180$ through $t_0 - 15$ calibration slice;
5. requires at least two available donors and 30 retained observations in each required residual window;
6. records an input-window failure if those calculation conditions fail;
7. computes diagnostic intervals, placebos, donor sensitivity, and protocol evidence tiers for complete cases; and
8. preserves complete cases, failures, and unavailable diagnostics.

This is an implementation specification for a metadata-anchored audit; it does not identify a physical mechanism.

Certificate behavior on held-out pairs

Group	Pair coverage	Observed scope error	Envelope violations	Oracle recovery
Overall	39.1%	0 observed errors	0 observed errors	97.7%
$q = 0\%$	0%	--	0 observed errors	--
$q = 25\%$	10.9%	0 observed errors	0 observed errors	87.5%
$q = 50\%$	38.3%	0 observed errors	0 observed errors	94.2%
$q = 75\%$	66.4%	0 observed errors	0 observed errors	98.8%
$q = 100\%$	79.7%	0 observed errors	0 observed errors	100%

Figure 13: Supplementary certificate-validity summary. “0 observed errors” denotes a finite-grid observation under the bounded generator, not a population-risk estimate.

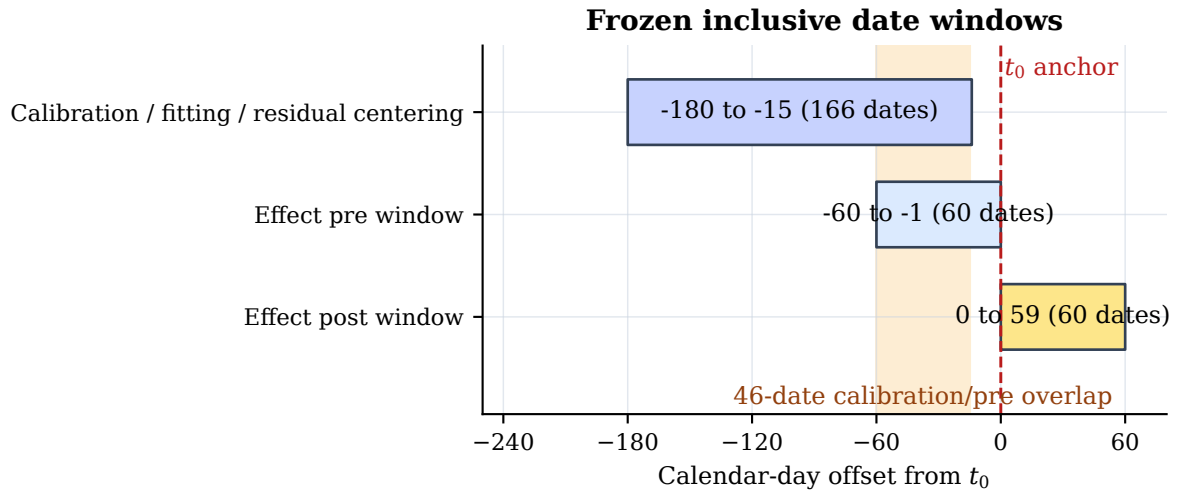


Figure 14: Calibration and displayed pre-anchor windows in the AQS audit. They overlap by 46 calendar dates, so displayed pre-anchor residuals are not an independent validation set. No post-anchor outcome enters fitting.

Deployment residual. For a qualified target $i = (s, p)$ and donors $j \in N_a$, the AQS audit uses

$$r_{i,t} = z_{i,t} - \sum_{j \in N_a} w_{ij} z_{j,t} - \text{median}_{u \in T_{\text{cal}}} \left(z_{i,u} - \sum_{j \in N_a} w_{ij} z_{j,u} \right).$$

Its displayed effect is the post-minus-pre median residual on the log scale. It is a diagnostic local residual discontinuity, not a causal physical-bias estimate or a corrected concentration.

A.4 Representative AQS cases

A.4.1 Selection policy and source provenance

Examples can make an audit readable, but selected examples can also conceal a failure mode or overstate a method’s scope. The three cases in this section are therefore not hand-picked for their visual contrast or effect magnitude. The display configuration selects one complete supported-candidate anchor and one complete not-supported anchor by minimum distance to the within-tier median standardized score, with anchor-ID tie-breaking. It selects the lexicographically first donor-insufficient inconclusive anchor to make the absence of a qualified counterfactual visible.

Table 20: Predeclared effect-window and reporting-scale sensitivity.

MetaShift window	Complete events	Median log effect	Sign agreement with 60 days
45 days	224	-0.08769	93.3%
60 days	228	-0.08704	reference
90 days	196	-0.08064	93.9%

Method	Log/raw sign agreement	Spearman ρ
MetaShift fixed	95.2%	0.829
Standard SC	92.5%	0.868
Nearest-neighbor DiD	92.1%	0.843

Note. Window sensitivity adds a full method-stability check for every target and donor window. Reporting-scale agreement is a concordance diagnostic and does not equate raw and log causal estimands.

Table 21: One-factor screening sensitivity with three required distinct donors.

Setting	Eligible before donor rule	With at least 3 donors
Primary (100 km)	563	238
Coverage 70%	563	243
Coverage 80%	563	232
Stable window 45 days	572	252
Stable window 90 days	543	214
Transition gap 3 days	512	212
Transition gap 14 days	589	254
Donor radius 50 km	563	102
Donor radius 200 km	563	426
Correlation $\rho \geq 0.50$	563	241
Correlation $\rho \geq 0.70$	563	230

Note. One predeclared setting changes at a time. These counts describe audit availability, not detected-effect frequency.

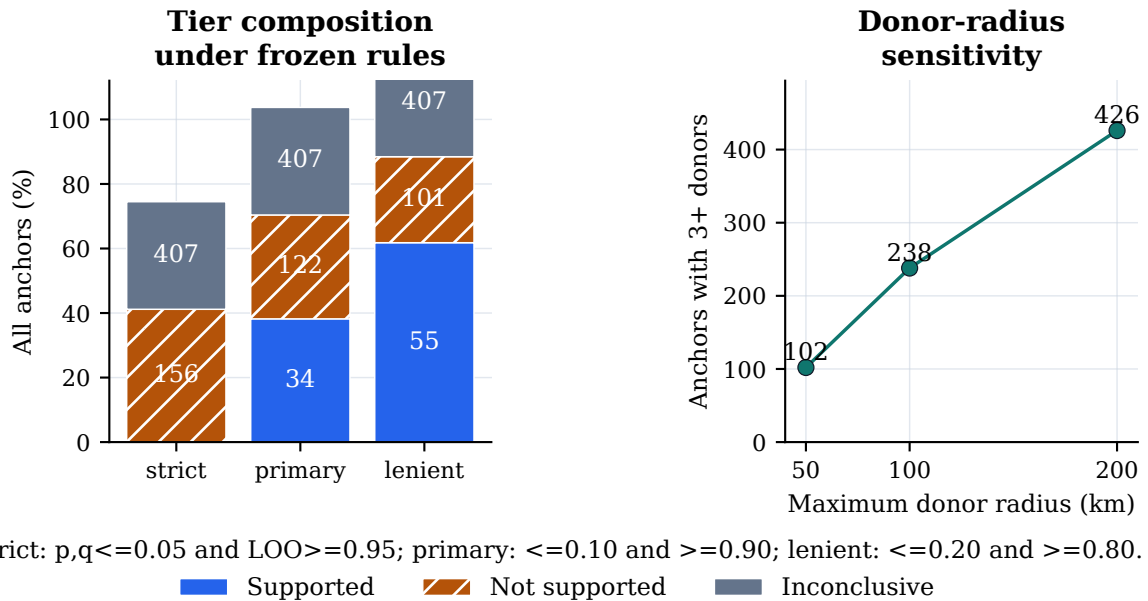


Figure 15: Predeclared evidence-tier and geographic-screening sensitivity. Changes describe availability and rule dependence, not physical causality or a search for a favorable specification.

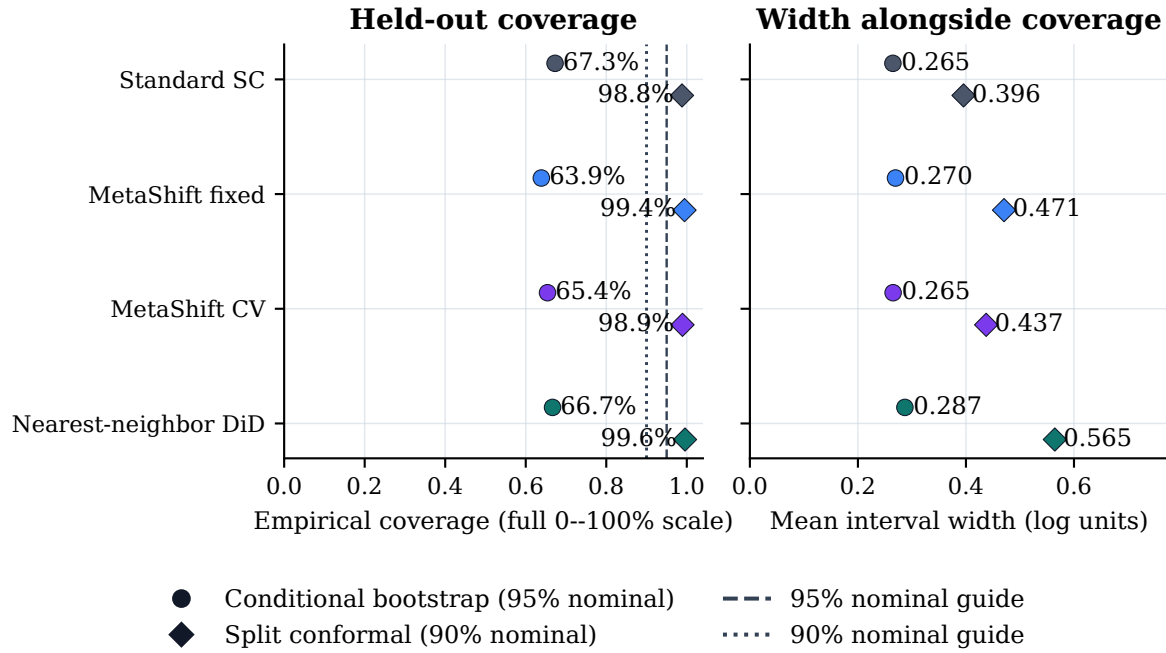
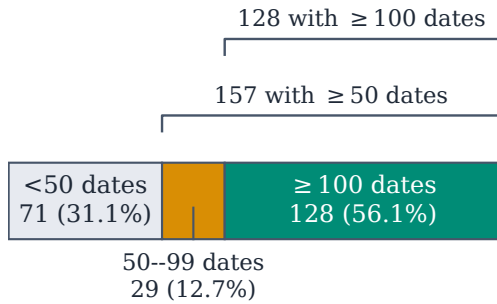


Figure 16: Supplementary empirical held-out synthetic interval coverage and mean width. Fixed conditional intervals under-cover nominal 95% coverage, whereas split-conformal intervals over-cover nominal 90% coverage.

(a) Placebo-date availability



(b) Raw probability, n=157

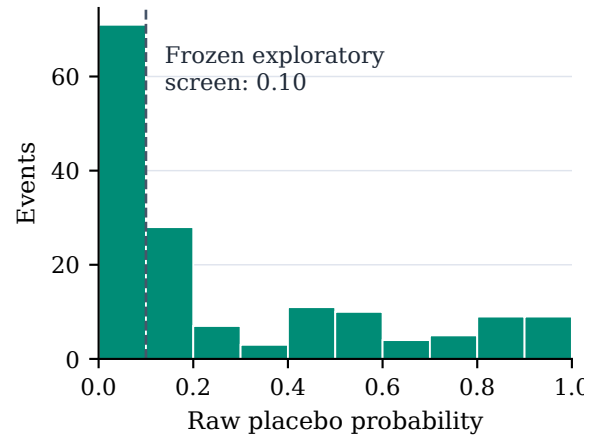
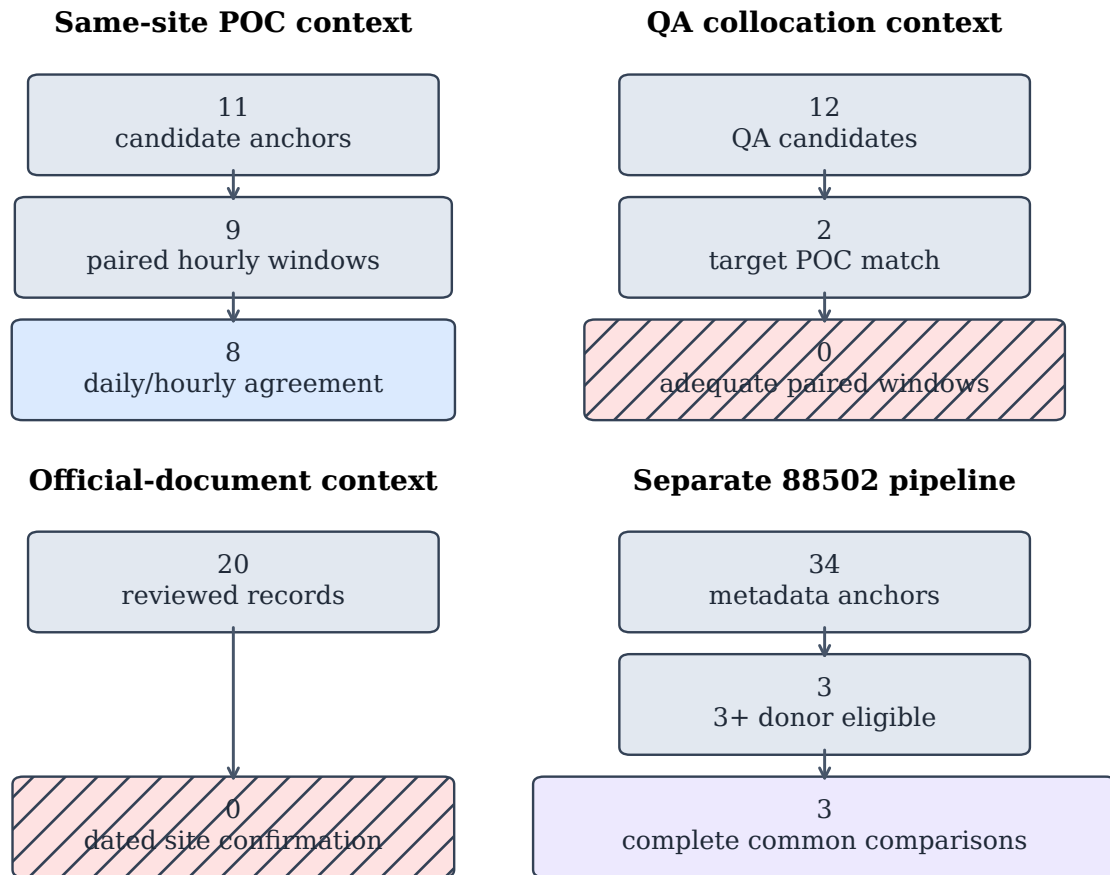


Figure 17: Supplementary time-placebo availability and raw within-event placebo-score distribution. These are diagnostic falsification quantities, not causal probabilities.

Table 22: Claim boundaries imposed by the MetaShift-Bench protocol.

The evidence can support	The evidence does not establish	Required interpretation
A reproducible metadata-anchored local residual audit	That a Method Code transition is a physical replacement or fault	Treat the transition as an anchor, not physical ground truth.
Benchmark-specific synthetic trade-offs among frozen methods	A robust aggregate MetaShift superiority claim	Report the paired intervals and retain the negative result.
Protocol-defined evidence tiers and abstentions	Instrument bias, true pollution correction, or causal attribution	Use tiers only to prioritize further records-based review.
Conditional interval behavior on frozen synthetic data	Coverage-calibrated confidence intervals for real-event physical bias	Describe real-event intervals as diagnostic.

Note. These boundaries are design constraints, not post hoc caveats.



All pathways are contextual evidence or feasibility checks.
None establishes a physical cause of a metadata transition.

Figure 18: Supplementary contextual-evidence ladders for same-site POC, QA-collocation, targeted document review, and the separate 88502 pipeline. Unavailable evidence remains visible.

Table 23: Deterministically selected representative audit cases.

Case	Metadata anchor	Disposition	Saved diagnostic summary
Supported candi- date	12-057-0113 poc1-2023-12-01	Complete	Effect -0.12777; fixed [-0.17258, -0.10511]; nested [-0.17399, -0.10560]
Not supported	06-031-0004 poc8-2021-01-12	Complete	Effect -0.08240; fixed [-0.21609, 0.03012]; nested [-0.24309, 0.03662]
Inconclusive	01-103-0011 poc3-2023-02-01	Donor insufficient	No counterfactual, effect, or interval is imputed.

Note. Supported and not-supported cases minimize absolute distance to their within-tier median standardized score, with anchor ID tie-breaking. The inconclusive case is the lexicographically first donor-insufficient anchor and deliberately has no counterfactual or interval. Intervals are diagnostic, not calibrated physical-bias confidence intervals.

The configuration, selected donors, archive manifest checksum, donor-table checksum, fitted weights, and reconstruction error are stored in the generated case-study manifest.

For the two complete cases, the renderer reads only the checksum-pinned public EPA archives, applies the frozen cleaning rule, reconstructs fixed-weight MetaShift with the reliability-prior penalty from pre-anchor data, and fails if the reconstructed saved log effect differs by more than 10^{-9} . The raw archives remain outside version control. The third case intentionally has no reconstructed counterfactual: its audit disposition is that fewer than three qualified distinct physical donors were available.

A.4.2 Interpretation boundaries

The supported-candidate panel illustrates a case in which the predeclared diagnostic conditions jointly hold. That status prioritizes the anchor for further station-history and technical-record review; it does not identify a device, a maintenance action, a calibration change, or a measurement bias. The not-supported panel illustrates the opposite distinction: a visible target series and a complete counterfactual are not sufficient when available diagnostics fail the frozen evidence rules. It would be misleading to discard that case merely because its residual plot is less dramatic.

The inconclusive panel is equally important. A target series may have an apparent before/after pattern, yet the protocol cannot form the common cross-site comparison if the distinct-donor requirement fails. MetaShift-Bench therefore abstains rather than borrowing a same-site POC, relaxing the donor rule after inspection, or assigning a physical explanation. This case makes the audit’s coverage boundary concrete: lack of a qualified comparison is lack of evidence for this protocol, not evidence that nothing happened.

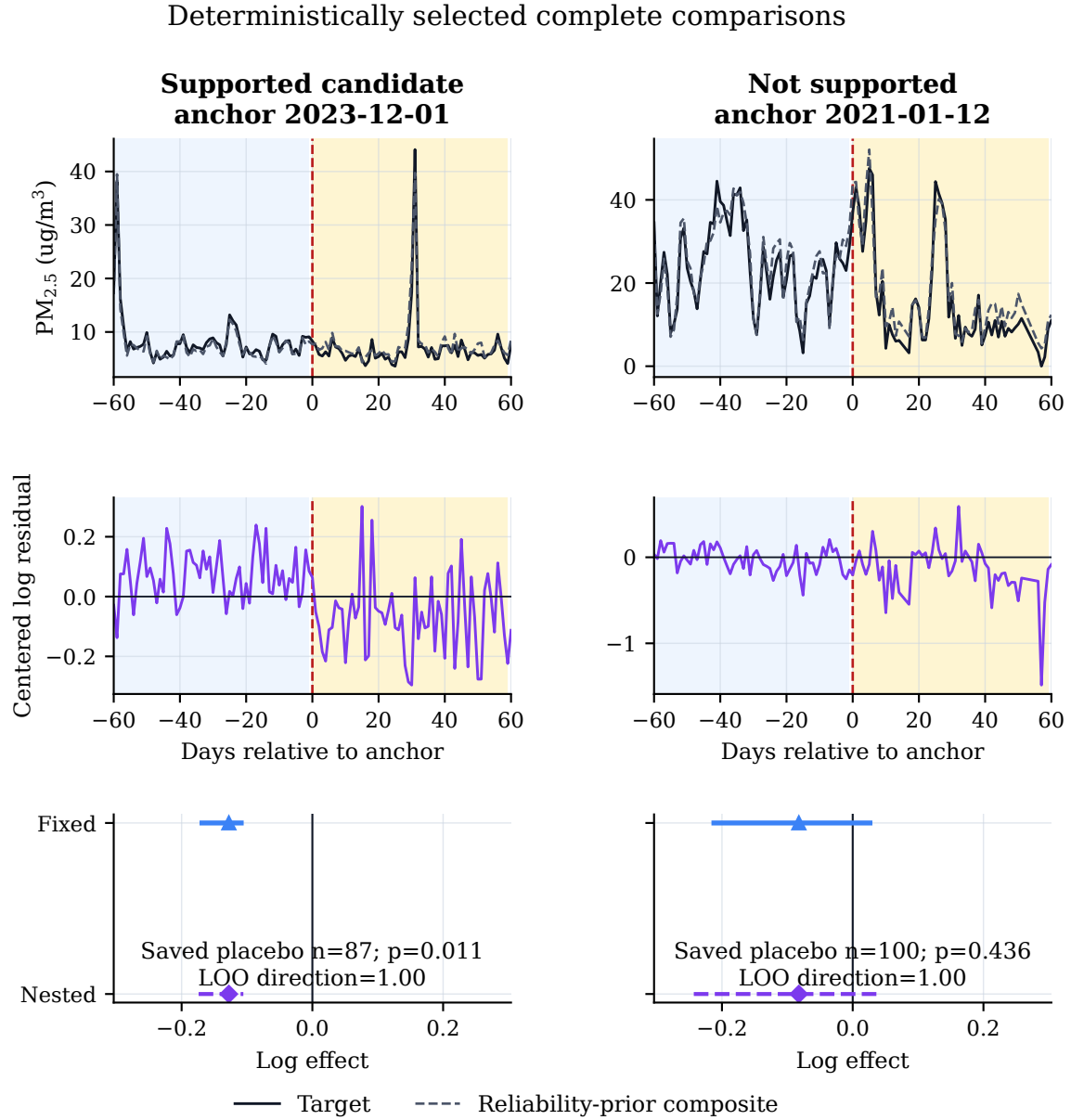


Figure 19a: Deterministically selected complete representative cases. The top row shows retained daily target values and the pre-anchor-fitted reliability-prior MetaShift counterfactual in $\mu\text{g}/\text{m}^3$; the middle row shows calibrated log residuals; and the lower row shows saved placebo summaries with fixed and nested diagnostic intervals. Blue and amber shading mark the 60-day pre- and post-anchor comparison windows. The figure displays saved diagnostic outputs, not evidence of a physical instrument event.

Deterministic abstention example: no counterfactual is imputed

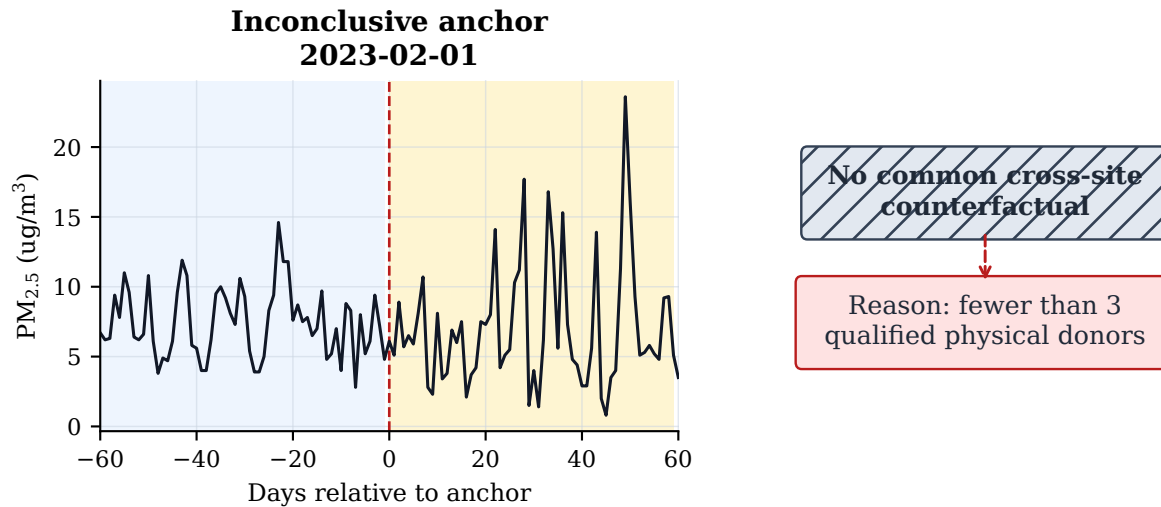


Figure 19b: Deterministic representative audit abstention. The retained target series is shown without a constructed counterfactual because fewer than three qualified distinct physical donors were available. The protocol therefore records an unavailable common comparison and does not impute a placebo, effect, or interval.

A.5 Applicability and anchor concentration

The frozen inventory has 393 of 563 anchors (69.8%) dated in 2023. Two reported code-pair transitions, 236 to 636 and 238 to 638, account for 354 of these 393 anchors (90.1%). This concentration is descriptive only: a code pair, time cluster, or method-context document does not establish an administrative, software, hardware, policy, calibration, or data-processing cause for an individual site record.

Table 24: Appendix-only temporal concentration of reported 88101 metadata anchors.

Old code	New code	Abbreviated reported metadata change	Count
236	636	T640, baseline reported name -> T640, alignment-enabled	234
238	638	T640X, baseline reported name -> T640X, alignment-enabled	120

Note. Of 393 2023 anchors, these two pairs account for 354 (90.1%). The largest date is 2023-07-01 (50), and the largest state-code subtotal is 42 (42). Descriptions abbreviate reported AQS Method Name strings; they do not establish a cause.

Evidence state	Protocol output	Claim boundary
Donor unavailable n=325	Abstain	No scope result; no mechanism claim
Input unavailable n=10	Abstain	No effect estimate; no mechanism claim
Complete comparison n=228	34 supported 122 not supported 72 inconclusive	No verified failure or causal bias; no automatic correction
Mechanism claims always require station histories, technical records, and human review.		

Figure 20: Claim-boundary matrix for the AQS audit. Rows partition all 563 primary anchors into donor-insufficient, input-window-failure, and complete-comparison states; the complete row separately reports observational evidence tiers. Every row excludes a physical-mechanism conclusion.

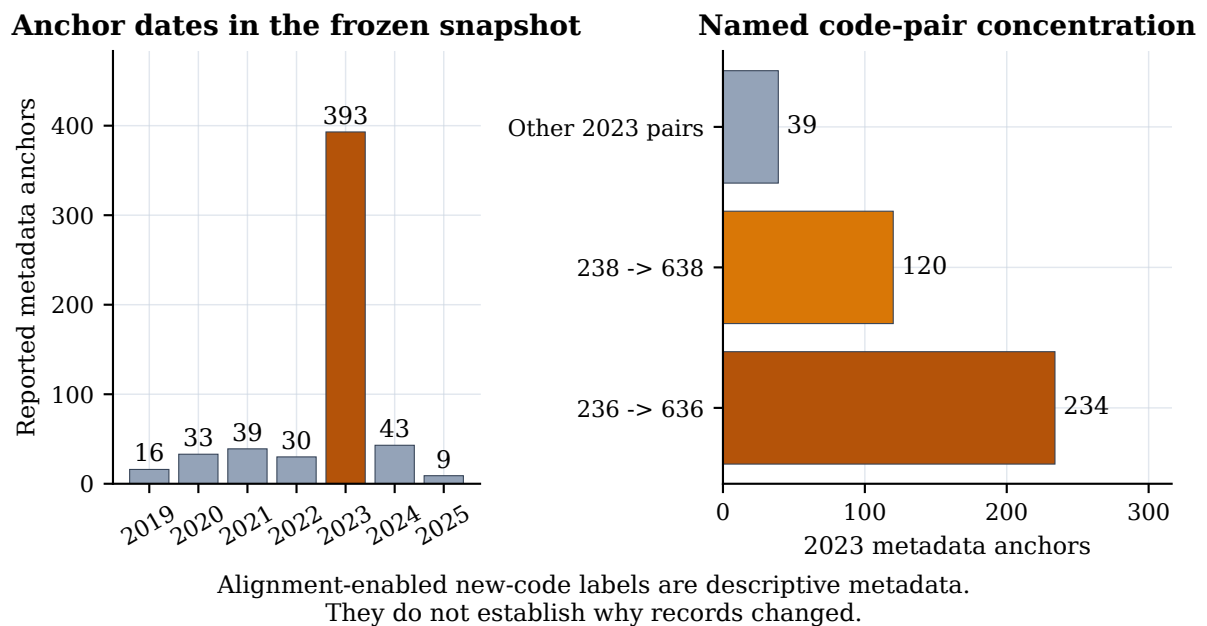
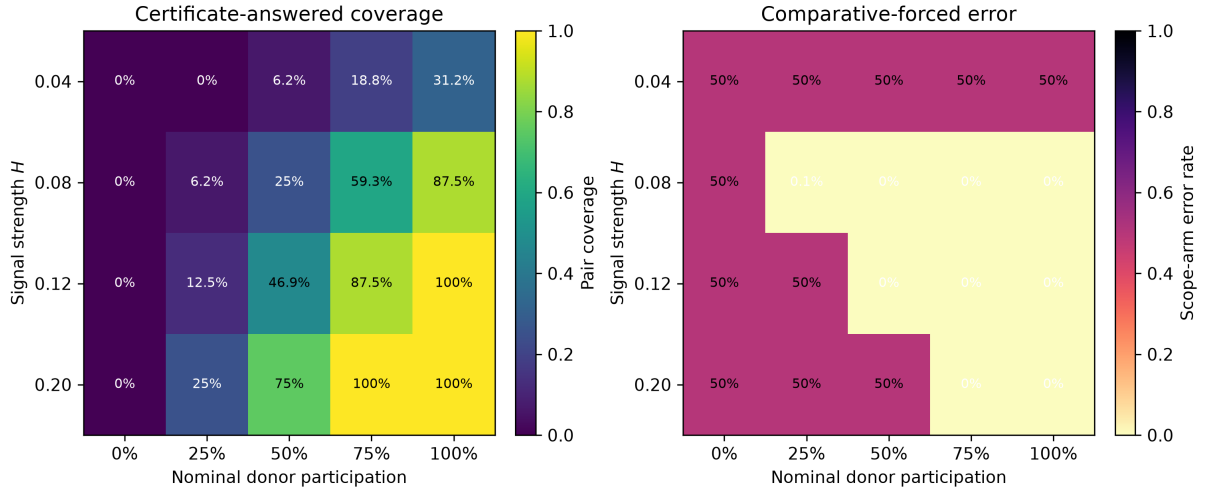


Figure 21: Descriptive concentration of reported 88101 metadata anchors. The association between code-pair labels and dates is not evidence of cause.

Table 25: Classification and boundary of the principal claims.

Statement	Status	Boundary
Information refinement and risk-coverage principles	Established theory	Cited background, not a MetaShift-Bench invention.
Target-only impossibility under the stipulated paired law	Established theory specialized to this construction	Requires balanced matched target laws and no target-side leakage.
Availability-normalized partial-scope identity and interval separation	New construction-specific derivation	Requires analysis-scale intervention and stated finite envelopes.
Held-out frontiers, gain, and certificate behavior	Frozen empirical observation	Applies only to the finite predeclared policies and generator.
Real AQS scope accuracy or physical mechanism	Explicit nonclaim	Requires independent labels, records, and human review.

**Figure 22:** Supplementary failure-mode map. Low signal and adverse comparison conditions remain visible as forced-comparative errors or certificate abstentions; $q = 0$ is an observational-identity negative control.

A.6 Claim classification and threats to validity

Table 26: Threats to validity, current boundary, and needed future evidence.

Validity class	Current boundary	Needed future evidence
Construct	Method Code and POC records are anchors, not physical device, calibration, siting, or maintenance ground truth.	Dated station histories, calibration records, deployment logs, and independently reviewed taxonomy.
Synthetic scope	Target-fixed labels and bounded panels may not represent operational networks; certificate inputs are known only to the generator.	Independent blind generators, naturally occurring labeled scope events, and admissible external records.
Internal	Weather, emissions, processing, and donor instability can make a target-reference residual; diagnostics constrain but do not identify cause.	Meteorological covariates, emissions records, network operations logs, and prospective comparator studies.
Statistical	Selection, serial dependence, multiplicity, and incomplete resampling limit real-event interval interpretation.	A computationally feasible predeclared full selection-aware coverage study on a new blind set.
External	The U.S. PM _{2.5} archive, availability rules, radius, and intervention families limit transfer.	Independent footprint-disjoint benchmarks from other networks, eras, and pollutant parameters.

B Reproducibility package

The project preserves two immutable evidence streams. The stable-regime benchmark and AQS deployment audit are bound to:

v0.3.2-evidence-final
57d678ecabebff724d898abe626c9ef80538775b

Their manifest records the source artifacts, checksums, and generated presentation assets. The target-fixed answerability experiment is separately bound to:

v0.5.0-answerability-freeze
14fd0fee4fb015e6c661299041e35ff704a27286
receipt SHA-256 prefix 954fc9b56a8f5266...

Its result verifier checks source and protocol bindings, full-grid accounting, target and $q = 0$ identity, split separation, and deterministic result consistency without rerunning the executor. The execution authority is recorded under `v0.5.0-answerability-execution-claim`.

The repository distinguishes archived-result verification from third-party reproduction. Read-only verification confirms that the preserved artifacts and their claimed provenance match. Reproduction on another environment requires the pinned requirements and records comparison; the frozen two-environment check applies to the designated core artifacts, not every local file, timestamp, cache, or renderer output.

The report build regenerates presentation-only tables and figures from verified inputs, validates source structure, references, numeric claims, and asset determinism, then performs a clean LaTeX build, font audit, and visual review. It never changes the archived evidence. The one-time target-fixed executor is not part of the document-build path and must not be rerun to improve or replace its results.

Table 27: Frozen evidence release verification record.

Record	Value
Evidence tag	v0.3.2-evidence-final
Frozen evidence commit	57d678ecabeb
Synthetic result label	stable_full_v2
Case manifest SHA-256	065b1b65c231c529...
Release-gate checks passed	35/35
Document-consistency checks passed	12/12
Markdown number checks passed	57/57
Two-environment core hashes matched	34

Note. The two-environment statement applies only to the listed designated core artifacts in the frozen reproducibility comparison, not to every local file.

C Acknowledgements, contribution statement, and required disclosures

HUMAN COMPLETION REQUIRED. This section is an intentionally uncompleted truthful-disclosure template. It must be replaced by a 500–1500 word account, verified by every student author and supervising teacher, before the report is submitted. Do not state that a person performed work they did not perform. Do not state that an advisor, school, organization, or AI system did or did not contribute unless the authors have verified the statement. Before submission, all student authors and supervising teachers must verify the exact final PDF and complete the required academic-integrity declaration, signatures, institutional stamps, and final truthfulness attestation.

Topic origin

HUMAN COMPLETION REQUIRED: Explain how the team selected the question, including the intellectual path from the original observation or reading to the metadata-anchor audit formulation. State whether a supervising teacher or other person suggested the topic and how the students refined the question.

Data acquisition and analysis

HUMAN COMPLETION REQUIRED: Identify who downloaded or reconstructed the public EPA AirData inputs, who inspected data rules, who implemented or reviewed the cleaning and donor-uniqueness logic, and who ran or checked the frozen analyses. State only the actual use of public AQS bulk files, public API endpoints, and local credentials. Do not include credentials, raw API responses, or private information.

Experimental design and writing

HUMAN COMPLETION REQUIRED: State which student(s) designed the synthetic evaluation, maintained the no-tuning boundary, generated figures and tables, interpreted limitations, and drafted or revised each report section. Describe the physical-donor uniqueness defect and remediation only if the authors personally understand and verify that account.

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